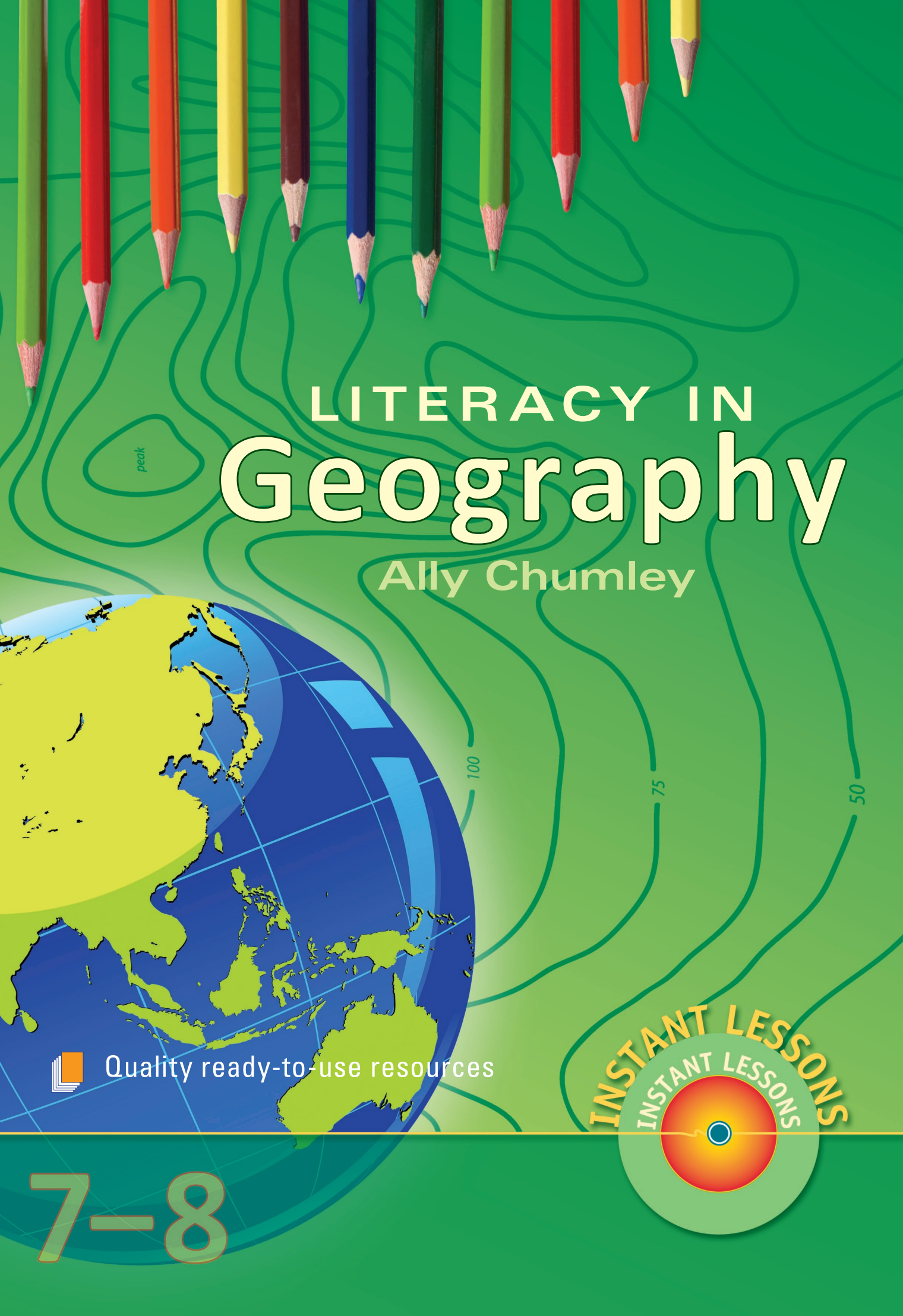




LITERACY IN Geography

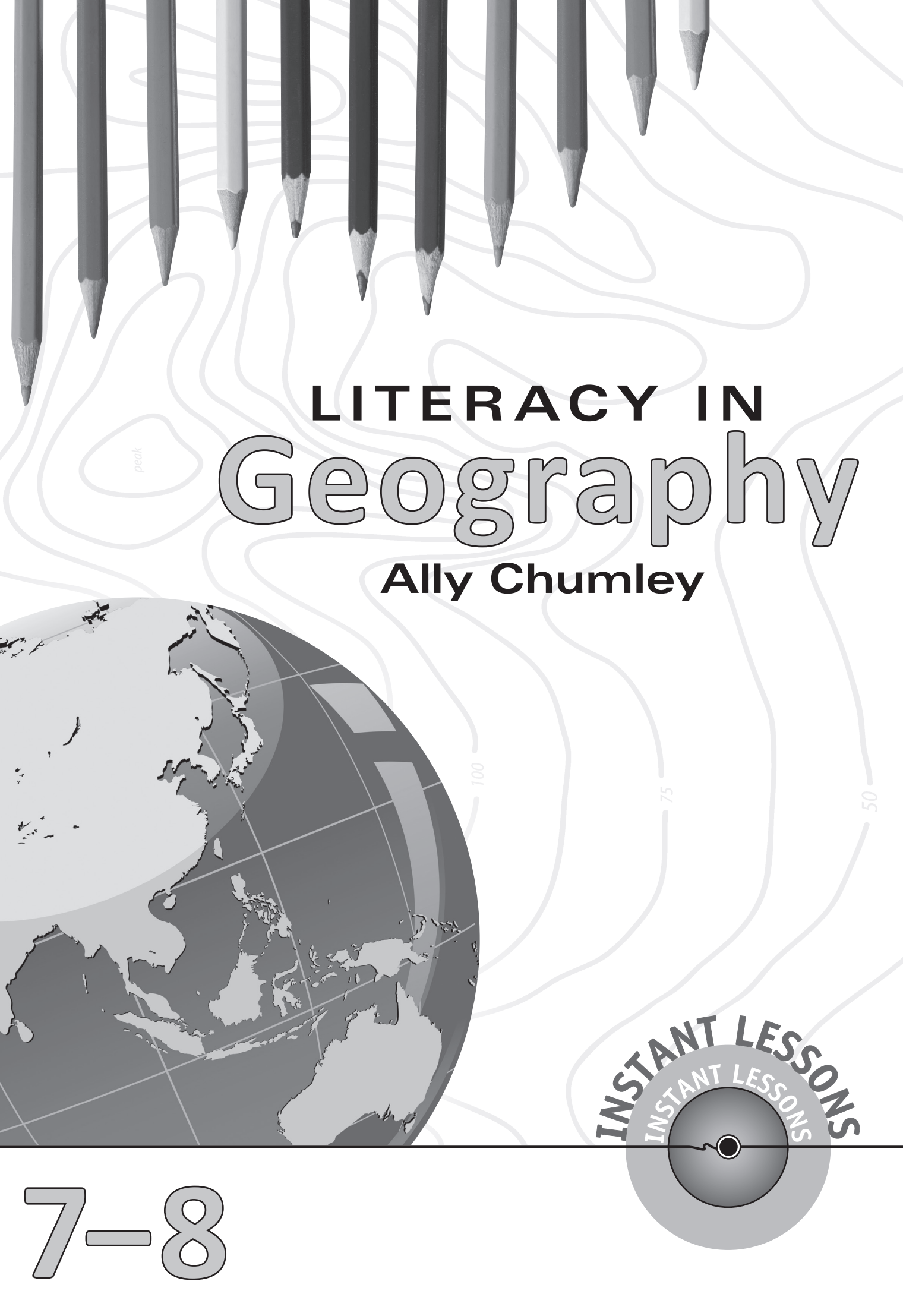
Ally Chumley



Quality ready-to-use resources

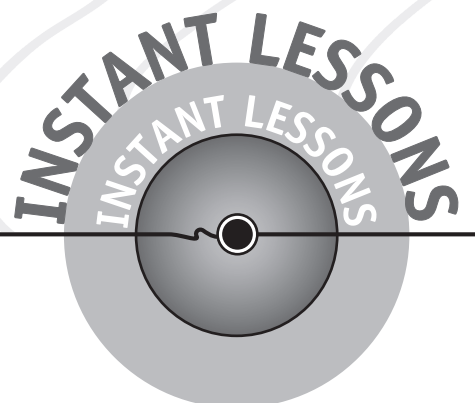
7-8



The background of the cover features a light gray topographic map with contour lines. At the top, several pencils of various shades of gray are arranged vertically, pointing downwards. The title 'LITERACY IN Geography' is centered, with 'LITERACY IN' in a bold, sans-serif font and 'Geography' in a large, outlined, sans-serif font. Below the title, the author's name 'Ally Chumley' is written in a bold, sans-serif font.

LITERACY IN Geography

Ally Chumley



7-8

Ally Chumley is a secondary teacher specialising in English, History and Geography. Her formal credentials include the degrees of Bachelor and Master of Education. Ally has written educational resources for teachers and learners across a range of subjects. She enjoys creating books that make difficult subjects accessible and inspiring to young people.

Ally loved going to school so much that she has never officially left. She boasts that her teaching career has made her Australia's foremost canteen food connoisseur. These days Ally lives on the mid-north coast of NSW with her husband and two boys. She has an enduring interest in reading, writing and educational research. She also wastes a lot of time watching rugby league, painting pictures and riding camels.



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Introduction

The *Literacy in Geography* series is designed to help students understand essential geographical terms. The work sheets in *Literacy in Geography* 7–8 revise and simplify core concepts, which can often present problems to otherwise enthusiastic geographers. *Literacy in Geography* 9–10 presents more challenging concepts and activities. Using these resources, teachers can support students to become comfortable with the vocabulary of geography, build skills in using an atlas, employ a range of research tools, and work with data and statistics. The work sheets can be used as complete lessons, or can be effectively incorporated into a lesson plan. We hope students enjoy completing them and reap great rewards through the study of our fascinating world and its people.

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Answers

46



Australia's natural environment

An island continent

1 Which statement is correct in each pair? Circle **TRUE** or **FALSE**.

- | | | | |
|----------|---|-------------|--------------|
| a | Australia is the world's largest continent and smallest island. | TRUE | FALSE |
| | Australia is the world's largest island and smallest continent. | TRUE | FALSE |
| b | Australia is mostly desert. | TRUE | FALSE |
| | Australia is mostly farmland. | TRUE | FALSE |
| c | Australia is densely populated, relative to its size. | TRUE | FALSE |
| | Australia is sparsely populated, relative to its size. | TRUE | FALSE |

Ocean, sea or desert?

2 Circle the correct term to complete these names for Australia's oceans, seas and deserts.

- | | | |
|---|--|--|
| the Pacific Ocean / Sea / Desert | the Strzelecki Ocean / Sea / Desert | the Coral Ocean / Sea / Desert |
| the Great Sandy Ocean / Sea / Desert | the Timor Ocean / Sea / Desert | the Gibson Ocean / Sea / Desert |
| the Tasman Ocean / Sea / Desert | the Tanami Ocean / Sea / Desert | the Arafura Ocean / Sea / Desert |
| the Simpson Ocean / Sea / Desert | the Indian Ocean / Sea / Desert | the Southern Ocean / Sea / Desert |

State or territory?

3 Which statement is correct in each pair? Circle **TRUE** or **FALSE**.

- | | | | |
|----------|--|-------------|--------------|
| a | Tasmania is a state. | TRUE | FALSE |
| | Tasmania is a territory. | TRUE | FALSE |
| b | The ACT is a territory within a state. | TRUE | FALSE |
| | The ACT is a state within a territory. | TRUE | FALSE |

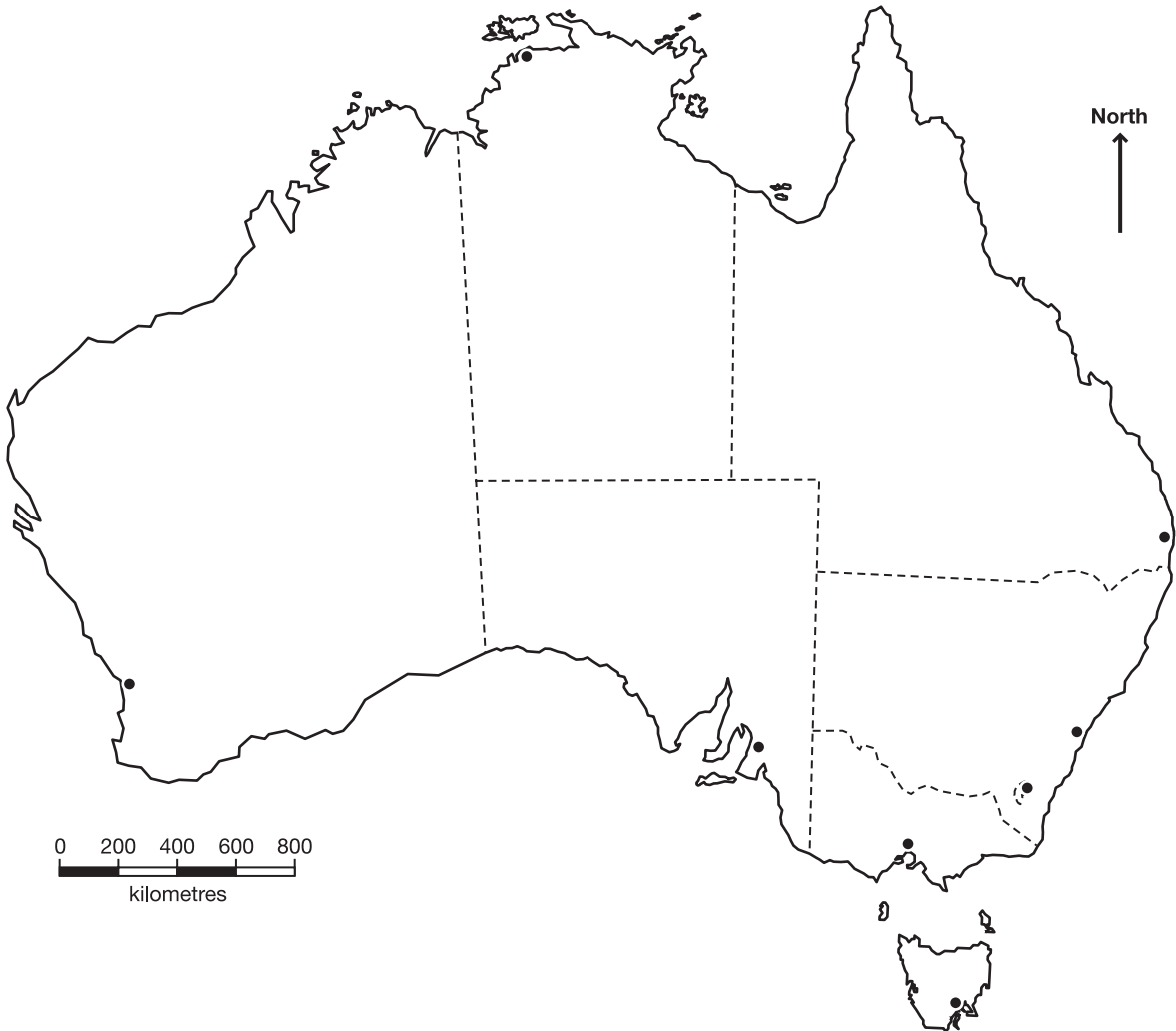
Australia's natural features

4 Choose the correct answer and write it in the space provided to complete each statement.

- a** The World Heritage site known as the Three Sisters is found in _____.
the Flinders Ranges / the Glasshouse Mountains / the Warrumbungles / the Blue Mountains / Kangaroo Valley
- b** Australia's largest river system forms _____.
the Murray-Darling Basin / the Darlington River / the Murchison River / the Victoria River / the Yarra River
- c** A star constellation visible from Australia's position in the Southern Hemisphere is _____.
the Union Jack / the Southern Cross / the Southern Lights / the Zodiac
- d** The world's largest sand island is found in Australia. It is called _____.
Australia Island / Bathurst Island / Melville Island / Lord Howe Island / Fraser Island
- e** Australia's highest mountain is part of the Great Dividing Range. It is called _____.
Mount Townsend / Rams Head / Mount Twynam / Alice Rawson Peak / Mount Kosciuszko

States and territories of Australia

1 On the map, write in the full names of the states and territories. Then label each state and territory’s capital city.



Estimating distances

2 Using the scale on the map above to estimate the distances, circle the correct answer in each pair.

- | | | | |
|--|---------|----|---------|
| a Darwin to the western-most point of the continent | 180 km | OR | 1100 km |
| b Melbourne to the northern-most point of the mainland | 3000 km | OR | 2100 km |
| c the eastern-most point of New South Wales to Hobart | 1680 km | OR | 700 km |

Marking population distribution

- 3 Colour or shade the areas of the map that are most densely populated.
- 4 What important geographical characteristics of the natural and human environments do these areas share that make them attractive to people?

Australian flora and fauna

1 Complete the alphabetical list of Australian flora and fauna (plants and animals) in the table below. There is more than one correct answer in each case.

	Flora	Fauna
A	acacia	Azure kingfisher
B		
C	coolibah tree	
D	dogwood	
E		
F	frangipani	fur seal
G		gecko
H	hibiscus	
I	iris	
J		jabiru
K	kangaroo paw	
L	lilly pilly	lorikeet
M		
N		numbat
O	orchid	osprey
P	prickly pear	
Q	quandong	quokka
R	red gum	
S	Sturt’s desert pea	sea lion
T		
U	umbrella tree	Ulysses butterfly
W		
Y	yacca	yabby

The term fauna applies to all living creatures, and includes mammals, reptiles, birds, fish and insects.

Describing flora

2 Choose TWO kinds of flora listed in the table at the left, and write descriptions of them. Draw sketches of each plant in the boxes provided.

Flora name: _____

Description: _____

Sketch:

Flora name: _____

Description: _____

Sketch:

marsupial

reptile

insect

bird

arachnid

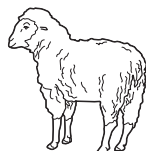
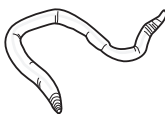
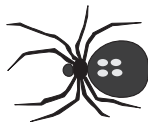
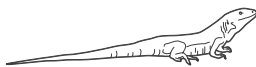
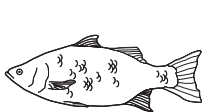
mammal

fish

invertebrate

Categories of fauna

3 Choose the correct fauna category from the box at the right, and write it under each picture.



Classifying wildlife

4 Sort these Australian flora and fauna into their correct categories by writing them in the table below.

bottlebrush	bandicoot	lace monitor	budgerigar	witchetty grub	catfish
ghost gum	dolphin	blackbutt	death adder	crow	mosquito
dugong	tortoise	currawong	cicada	lungfish	kelp
flying fox	skink	penguin	silver perch	camphor laurel	humpback
goanna	lyrebird	dragonfly	water hyacinth	koala	possum
Murray cod	turtle	crocodile	wombat	praying mantis	mango
barramundi	macadamia	cassowary	king brown	Bogong moth	Mangrove Jack
grasshopper	flathead	tawny frogmouth	emu	eel	ladybird

Plant	Mammal / Marsupial	Reptile	Bird	Insect	Fish

Sketching fauna

5 Make detailed diagrams of TWO kinds of native Australian fauna and label their important features.

Fauna name: _____

Fauna name: _____

Who eats who?

6 Draw a food chain diagram that incorporates at least FOUR types of Australian flora and fauna.

Food chain:

About Antarctica

- 1 Circle the most appropriate term from each pair in the passage.

HINT: Consider factual accuracy, shades of meaning and proper grammar.

Antarctica is situated in the **Arctic / Antarctic** Circle, and is surrounded by the **Northern / Southern** Ocean. It is the southernmost **country / continent**, and is the **location / place** of the South Pole. The Antarctic stretches over 14 million square **kilometres / metres**. About 98% of the land mass is covered by **ice / icicles**. The average **thickness / size** of the ice is 1.6 kilometres. Antarctica is the **warmest / coldest** continent and is very windy. It experiences very **little / small** precipitation as rain, so is technically classified as a desert. Plants that have **alternated / adapted** to the cold can survive there, with the main flora being species of moss, lichen and algae. The region is **population / populated** by penguins and other sea birds, seals, whales and fish. Few other creatures can survive the **frozen / freezing** temperatures.

Whale maths

- 2 Place these whale species in order of average length from longest to shortest.

HINT: Use the information provided in the table to convert the length of each species to metres first.

Whale species	Average length
Sperm	1500 cm
Fin	27 m
Southern Bottle Nose	9000 mm
Minke	1000 cm
Southern Right	17 m
Blue	0.03 km
Humpback	16 m

Whale species	Average length (longest to shortest)

- a If 25 Southern Right whales of average length formed a nose-to-tail chain, how long would the chain be in metres?

- b If one Sperm whale weighs 40 tonnes, on average, how many whales would be required to make a combined weight of 3000 tonnes? _____
- c A Sperm whale has an average weight of 40 tonnes. Sperm whales are only half as heavy as Fin whales. Humpbacks weigh 6 tonnes more than Sperm whales. What is the average total weight of two Sperm whales, two Fin whales and three Humpbacks? _____
- d Explain how you arrived at this figure. _____

The Antarctic food chain

- 3 In diagrams of food chains, the point of each linking arrow is toward the organism eaten by its predator. Draw a food chain that incorporates these Antarctic species:

krill phytoplankton whales

Orcas (killer whales) are classified as dolphins, and are not technically a whale species.

Food chain:

The physical and human environments

1 Geography is the study of local, national and global communities and environments. It examines groups of life forms including humans, animals and plants, and studies their interactions with the Earth’s natural resources. Decide whether each topic listed in the box involves the study of the **physical** or the **human** environment. Write the subjects in the correct columns.

determining the temperature inside an active volcano	measuring the height of the waves in an ocean tide
forecasting the weather	investigating unemployment in a particular city
investigating a factory’s waste disposal practices	discovering how a plant is cross-pollinated

Physical environment

Human environment

Key geographical questions

ISSUE: Many expensive fashion items are manufactured by people who receive unfairly low wages for their work.

2 Use the following prompts to write a set of questions a geographer might pose about this issue. You may need to do some research on the internet or in the library to get started.

What? _____

Where? _____

When? _____

How? _____

Why? _____

Is this a good or bad thing? _____

Can it be changed? _____

What can be done about it? _____

Tools geographers use

3 Choose THREE of the following tools used by geographers and explain what each one is used to investigate.

map	thermometer	photograph	survey	barometer	seismograph	tracking device
-----	-------------	------------	--------	-----------	-------------	-----------------

• _____ : _____

• _____ : _____

• _____ : _____

Geography mind map

4 In your notebook, create a mind map that summarises some of the areas of study covered by geography. Use these six headings: 1. Biological Elements 2. Physical Elements 3. Humans 4. Rocks, Minerals, Soils, Water, Sunlight, Air 5. Total Environment 6. Plants and Animals.

Weather words

1 Rearrange the letters in each of these words to form a term used in weather forecasting.

- a wells

b thorn
- c diet

d teas
- e stonk

f stew

Weather phrases

2 Rearrange the letters in each nonsense phrase to form a word or phrase related to the weather.
The first one has been done for you.

- a winch dill

wind chill

b little sea
- c coal sat

d ull potion
- e meet pure rat

f robe tamer
- g true messy press

h nux dive
- i heeting a few carrots

j peas mother

How we measure the weather

3 Complete the table showing units of measurement used in weather reporting.

Instrument	What is being measured	Unit	Symbol
barometer	atmospheric pressure	hectoPascal and millibar	hPa and mb
thermometer		degrees Celsius	
anemometer	nautical speed and wind speed		kn
ruler	length		m
hygrometer	relative humidity	per cent	
clock			min
seismograph		Richter local magnitude scale	M _L

Reporting the weather

4 Generate your own weather report by filling in missing words in the passage below. Base your report on today’s conditions in your region. Use your skills in estimation to invent realistic figures to add to your report.

It was a _____ day in _____ today. Cloud cover was _____.

Temperatures ranged from _____ degrees in the morning to _____ in the afternoon. Humidity was _____, and the UV Index reading was _____.

Fire danger was _____. Winds of up to _____ kilometres per hour were blowing from a _____ direction. Rainfall was _____. The Bureau of Meteorology tells us to expect _____ conditions tomorrow. Today’s temperature and rainfall figures around the district were:

Town / Area: _____

Average Temp: _____

Rainfall: _____

Town / Area: _____

Average Temp: _____

Rainfall: _____

Town / Area: _____

Average Temp: _____

Rainfall: _____

Weather word chains

5 Use these word chains to form separate weather terms. The letter ending each word is the first letter of the next. Write the words in the spaces provided.

a **BAROMETERAINFALLATITUDESWELL**

b **OUTLOOKNOTSOUTHAZEASTROUGHECTOPASCALS**

c **FROSTEMPERATUREARTHUMIDITY**

d **TROPICOASTALOWESTERLY**

e **FOGALEFORCEWINDEGREESCALECOLOGICAL**

f **METEOROLOGISTORNADOVERCAST**

g **FRONTIDESATELLITE**

h **POLLUTIONORTHWESTOTALFIREBAN**

Drawing weather symbols

6 Draw and label an example of each of the following weather symbols used in synoptic charts.

a high pressure system

a cold front

a tropical cyclone

Lines of latitude

1 Complete the passage below using words from the box.

line	North	centre	degrees	angle	zero	rotate	Equator	time
------	-------	--------	---------	-------	------	--------	---------	------

Imagine we could cut the Earth exactly in half along the theoretical line we call the E _ _ _ _ _ . The flat, cut surface is what we call the *Equatorial plane*. The middle of that plane is the Earth’s core and its angle is z _ _ _ degrees because it is the absolute c _ _ _ _ _ of the planet. To pinpoint a location on the Earth’s surface, we imagine drawing a l _ _ _ between zero and the point being described. This line creates an a _ _ _ _ of elevation above the Equator, or below it. These are what we call angles of *latitude*.

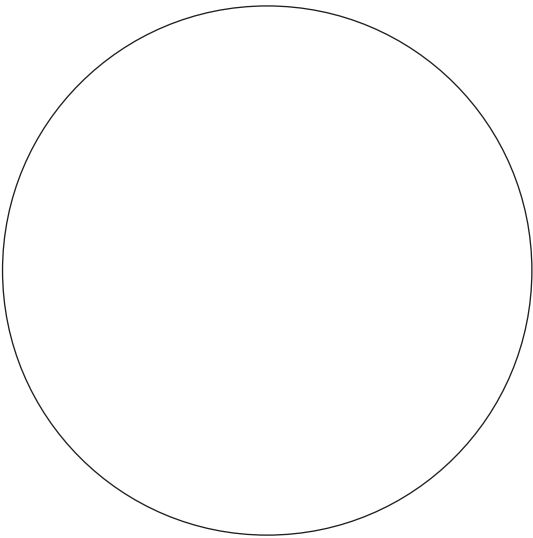
Because the Equator is a circle of 360 d _ _ _ _ _ , angles of latitude are measured in degrees of arc, minutes of arc and seconds of arc. These minutes and seconds are not the same as ordinary t _ _ _ measurements. They relate to the time it takes the Earth to r _ _ _ _ . The maximum latitude angle possible is 90 degrees north or south from the imaginary zero point at the Earth’s core. There are two such locations: the N _ _ _ _ Pole and the South Pole.

Drawing lines of latitude

2 Imagine this circle is the Earth. Sketch in some sample lines of latitude. Label the Equator on your drawing.

HINT: Lines of latitude are shown as circling the Earth horizontally, getting increasingly smaller as they get further away (both north and south) from the Equator.

The point where both latitude and longitude coordinates are zero is in the Gulf of Guinea (Atlantic Ocean) off the coast of West Africa. It is where the Greenwich Meridian line passes through the Equator at an angle of zero degrees.



Northern or southern hemisphere?

Australia	Greenland	South Africa	Canada	Finland	Peru	England	New Zealand
-----------	-----------	--------------	--------	---------	------	---------	-------------

3 Write the name of each country in the correct row to show whether it is in the northern or southern hemisphere.

Northern: _____

Southern: _____

Countries near the Equator

4 Look at an atlas or globe to complete the names of these countries that lie on or close to the Equator.

- a Central America: E _ _ _ _ _ r C _ _ _ _ _ a V _ _ _ _ _ a
- b Africa: K _ _ _ a U _ _ _ _ a Democratic Republic of C _ _ _ o
- c Asia-Pacific: B _ _ _ _ o M _ _ _ _ _ a I _ _ _ _ _ a

Lines of longitude

1 Complete the passage below using words from the box.

meridian	longitude	half	London	single	imaginary	curved
	reference		west	measured	circumference	coordinates

Angles of longitude are determined by the use of i_____ lines (or *meridians*) running from the North Pole to the South Pole. They start at a single point (the North Pole), bulge out around the Earth’s c_____ surface, then return to reach a s_____ point at the South Pole. A longitude number shows the point where a certain m_____ meets the Equator. Every meridian crosses the Equator, so the Equator is the point of r_____ for all longitude measurements. Zero degrees north / south and 90 degrees east / west are the c_____ of the North and South Poles. These lines of longitude are m_____ from the central line, which is located at Greenwich in L_____, England. The global standard of time measurement also has Greenwich as its base reference. The Greenwich Meridian is considered to represent zero degrees l_____. These angles are measured from zero to 180 degrees east or w_____ of the Greenwich Meridian. The location 180 degrees east of the Greenwich Meridian is the same as the location 180 degrees west, because 180 degrees is h_____ of the Equator’s total c_____ (360 degrees).

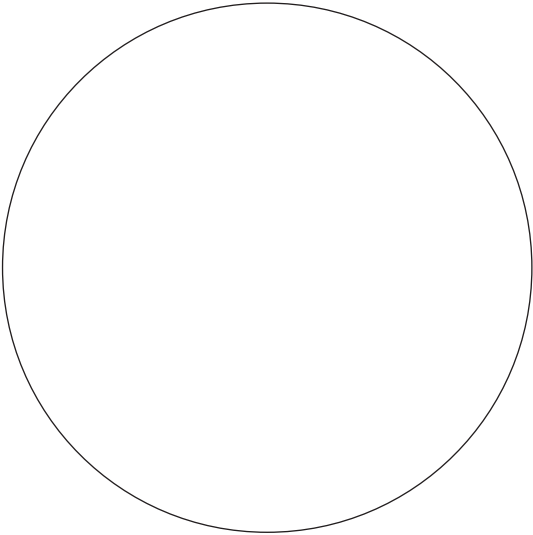
Drawing lines of longitude

2 Look at a globe or world map and sketch in the continent of Europe. Draw lines of longitude on the circle below. Label the Greenwich Meridian (in London, England).

HINT:

Lines of longitude look like the segments of a peeled orange, starting at a Pole at a single point then curving out over the Earth’s surface and meeting at a single point at the opposite Pole.

At the 180 degree line of longitude in the Pacific Ocean is a theoretical line, known as the **International Date Line**, that separates the two hemispheres. In order to accommodate the locations of various island countries, this imaginary line is not perfectly straight. Travelling east across the line requires you to subtract 24 hours from today’s date. To the left of the line, the date is always one full day ahead of the date for the locations to the right.



East or west?

3 Place a tick in the correct column to indicate the countries that lie within the Earth’s eastern and western regions, measuring from the Greenwich Meridian. Look at a globe or world map to help you.

Country	Eastern	Western
Morocco		
Egypt		
Argentina		
Turkey		
India		
Saudi Arabia		
Ireland		
Japan		
USA		
Austria		

GMT (Greenwich Mean Time) is now known as Coordinated Universal Time (abbreviated to UTC). Sometimes, it is written as GMT/UTC. UTC is the standard time zone from which we measure all the other time zones across the world.



Famous places around the world

The seven natural wonders of the world

1 Write the beginnings from the box next to the endings below to name the world’s top seven natural wonders.

Victoria	Paricutin	Grand	Aurora	Mount	Rio de	Great Barrier
----------	-----------	-------	--------	-------	--------	---------------

- a _____ Canyon in the United States of America
- b _____ Reef off the east coast of Australia
- c _____ Janeiro Harbour in Brazil
- d _____ Everest on the border between Nepal and Tibet
- e _____ Borealis visible from the Earth’s northern hemisphere
- f _____ Volcano in Mexico
- g _____ Falls on the border between Zambia and Zimbabwe

Rivers of the world

2 Write the name of each river listed in the box next to the correct description below.

Shannon	Amazon	Rhine	Nile	Seine	St Lawrence	Volga	Rio Grande
Colorado	Jordan	Zambezi	Yangtze	Thames	Ganges	Murray-Darling	

- a _____ This South American waterway is the world’s second longest river.
- b _____ This river runs through the Grand Canyon.
- c _____ Australia’s largest river system.
- d _____ A sacred river to Hindus.
- e _____ An ancient Near-Eastern river often mentioned in the Bible, which still flows today.
- f _____ The world’s longest river.
- g _____ Europe’s most important inland waterway.
- h _____ Its name means “great river” and it forms the border between the USA and Mexico.
- i _____ This river joins the North American Great Lakes with the Atlantic Ocean.
- j _____ This river flows through the city of Paris.
- k _____ The longest river in Ireland.
- l _____ This is the longest and most important waterway in England.
- m _____ Found in Russia, this is Europe’s longest river.
- n _____ China’s longest river.
- o _____ This African river incorporates the Victoria Falls.

Famous lakes

- 3
- Use an atlas to find the countries in which these famous lakes are found.
- a

Lake Geneva

b

Loch Ness
- c

Lake Eyre

d

Lake Baikal
- 4
- In the atlas, find the lakes that exist between these countries.

HINT: The last one is called a 'sea', although it is officially considered to be a lake.

- a

United States of America and Canada
- b

Tanzania, Uganda and Kenya
- c

Malawi, Tanzania and Mozambique
- d

Azerbaijan, Iran, Kazakhstan, Russia and Turkmenistan

Famous sites around the world

- 5
- Circle the site in each set that is not located in the same country as the others. Use an atlas or an online reference tool to help you.
- a

White Cliffs of Dover

Uluru

Wave Rock

the Devil's Marbles

the Three Sisters
- b

Lake Michigan

Lake Victoria

Lake Huron

Lake Ontario

Lake Superior
- c

London

Birmingham

Manchester

Dublin

Oxford
- d

the Seine

Marseilles

Paris

Calais

Kyoto
- e

the Valley of the Kings

the Nile

Mount Vesuvius

the Sphinx

the Great Pyramid
- f

Moscow

St Petersburg

Red Square

Berlin

Siberia
- g

the Great Wall of China

Beijing

the Yellow River

Johannesburg

the Yangtze River

Describing famous sites

- 6
- Choose TWO of the sites listed above and write descriptions of them using the plans below.

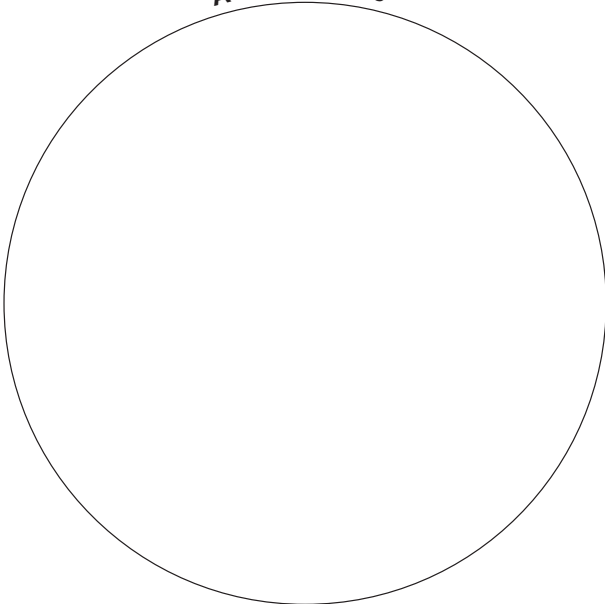
Site 1	Site 2
Country	Country
Type of site	Type of site
Physical characteristics	Physical characteristics
Reason for being famous	Reason for being famous
How you first heard of it	How you first heard of it

Which sphere?

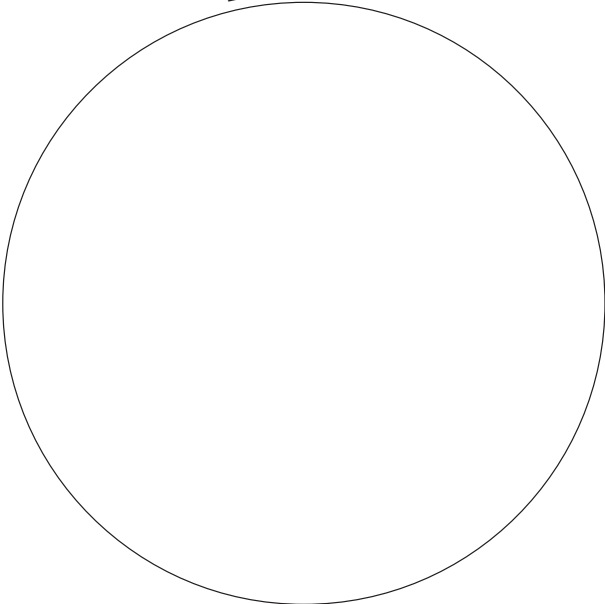
1 Write each of these terms in the correct geographical sphere.

crust	lichen	gases	rainforest	wind	rivers	consumer
precipitation	mountains	cirrus clouds	estuary	fault	deserts	waves
habitats	tsunami	oceans	topography	predator	plate tectonics	biodiversity
air	food webs	temperature	soil	meteorology	tides	snow
reproduction	mantle	hail	photosynthesis	ecosystem	ozone	geology
carbon dioxide	air pressure	valleys	stratosphere	plateau	vegetation	oxygen
dew	icebergs					

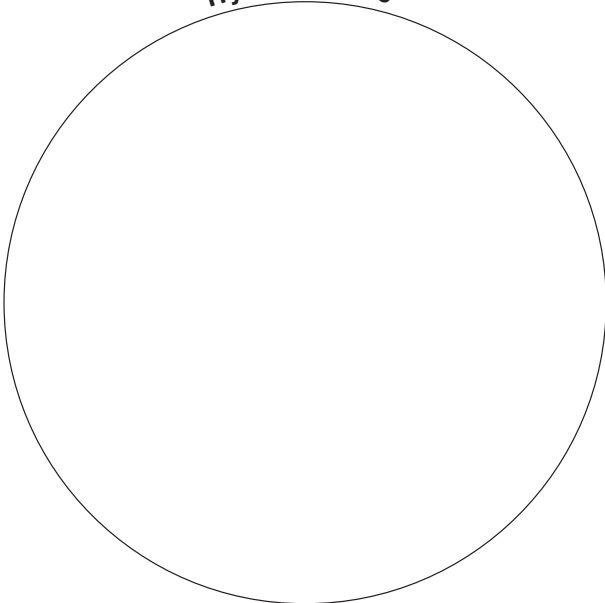
Atmosphere



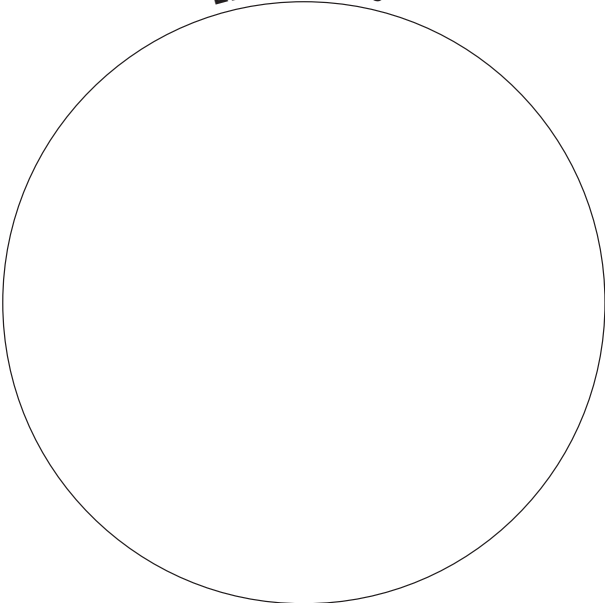
Biosphere



Hydrosphere



Lithosphere



Disasters of the spheres

2 Write the correct letter (A, B, H or L) next to each phenomenon to indicate the sphere that it most affects.

- | | | | |
|----------------------------------|------------------------------|--------------------------|--------------------------------|
| a tornado _____ | b air pollution _____ | c landslide _____ | d extinction _____ |
| e volcanic eruption _____ | f acid rain _____ | g cyclone _____ | h siltation _____ |
| i erosion _____ | j earthquake _____ | k disease _____ | l ocean pollution _____ |

Greek word origins

- 3 The meanings of many geographical terms can be worked out from their Greek origins. Choose the correct word to complete each sentence.
- a The prefix *atmos* means vapour or gas, so the *atmosphere* is the layer of _____ between Earth and space. **gases / gaseous / gassed**
 - b The prefix *bio* means life, so the term *biosphere* refers to all _____ things (plants, animals and humans and their habitats). **lively / living / alive**
 - c The prefix *hydro* means water, so the term *hydrosphere* refers to all of Earth's _____ resources and processes. **watery / water / wet**
 - d The prefix *litho* means rocky, so the term *lithosphere* relates to the Earth's _____, mantle, minerals and landforms. **crustal / crustaceans / crust**

Spelling shortcuts

- 4 Look at the names of the four spheres and answer these questions:
- a How many letters do you need to change in order to make *lithosphere* into *hydrosphere*? _____
 - b How many letters do *biosphere* and *atmosphere* have in common? _____
 - c Which of the names of the four spheres contains the fewest vowels? _____

Letter exchange

- 5 Follow the steps to make new words.

- a Look at the word ATMOSPHERE.

Remove the last five letters: _ _ _ _ _

Remove the second letter. What man's name do the remaining letters spell? _ _ _ _

Add letters T, Q, U, O and I: _ _ _ _ _

Remove the letter A: _ _ _ _ _

Now, arrange all the letters to spell the name of an insect that travels in the atmosphere:

_ _ _ _ _

- c Look at the word HYDROSPHERE.

Remove the first and second vowels and the letter S:

_ _ _ _ _

Add the word 'ATE' at the end of the remaining letters: _ _ _ _ _

Remove P and H: _ _ _ _ _

Remove another two letters to form a word that means 'add water to': _ _ _ _ _

Now, turn this word into a noun by removing E and adding the ending 'ION': _ _ _ _ _

Finally, add two letters to the start to make the word mean the opposite of 'having enough water':

_ _ _ _ _

- b Look at the word BIOSPHERE.

Replace the letter H with F: _ _ _ _ _

Remove all the letters that form the word 'SIP':

_ _ _ _ _

Remove the letters O and R: _ _ _ _

Rearrange the remaining letters to name a food product we get from one kind of animal in the biosphere: _ _ _ _

- d Look at the word LITHOSPHERE.

Remove the letter P: _ _ _ _ _

Spell two separate words using the remaining letters. Use the definitions below to help you.

A region made up of a number of towns:

_ _ _ _ _

A place to stay when you're away from home:

_ _ _ _ _

Describing natural phenomena

1 Unscramble the words and add the missing vowels to their synonyms (words with similar meaning). Then complete the table by naming the natural element (air, earth, fire, water) associated with each phenomenon.

Scrambled word	Unscrambled word	Synonym	Natural element
doornat		t w _ s t _ r	
sandildle		l _ n d s l _ p	
satumin		t _ d _ l w _ v _	
queerathak		t r _ m _ r	
cloycen		h _ r r _ c _ n _	
lizardbz		w h _ t _ _ _ t	
gof		m _ s t	
lodof		_ n _ n d _ t _ _ n	
mosstrand		w _ l l y w _ l l y	
ubifresh		b l _ z _	

Defining natural phenomena

- 2 Match the words to their definitions by inserting the correct letters in the spaces.
- a tornado _____ a rapid down-slope movement of land caused by water in the soil and rock
 - b landslide _____ a sudden movement of the Earth’s crust caused by friction between the continental plates
 - c tsunami _____ a heavy snowstorm combined with high speed wind
 - d earthquake _____ the overflow of water onto the land
 - e cyclone _____ an uncontrolled fire sweeping through a forest area
 - f blizzard _____ a large wave or series of waves caused by an under-sea earthquake
 - g fog _____ an extremely low pressure system resulting in destructive winds
 - h flood _____ a localised spiral of wind with extremely violent speed and powerful uplift
 - i sandstorm _____ ground-level condensation
 - j bushfire _____ fast-moving dust clouds, swept across the ground by high winds

Explaining natural phenomena

- 3 Choose ONE of the natural phenomena listed above and write a brief description of its causes and effects.
-

Writing definitions

4 Write your own definitions for these natural phenomena:

avalanche _____
thunderstorm _____
volcano _____

Coping with a natural disaster

5 Match each natural disaster to the correct emergency advice by writing the correct letters in the spaces.

- a tornado _____ Stay away from buildings. If stuck inside, take cover under something that won’t collapse.
- b earthquake _____ Stay inside, and try to park any cars undercover to avoid damage.
- c electrical storm _____ Make every effort to escape to high ground, and do it quickly.
- d flood _____ If driving, slow down and use special lights to enable you to see where you are going.
- e cyclone _____ Keep watch and try to stay out of its path. Take shelter underground if possible.
- f hail _____ Take shelter in a building and protect or reinforce glass windows and doors where possible.
- g fog _____ Take shelter indoors. If caught outside, stay low, moving away from any trees or tall objects.

Creating emergency action plans

6 An emergency action plan outlines the best way to avoid injury during an emergency situation. Plans include a short list of instructions about what to do and how to proceed to a safe location where you can be protected. Design a plan of action for the safety of the students and staff at your school for either a flood OR a fire.

Packing for an evacuation

7 Imagine you and your family are refugees from a war in your region. You have been asked to evacuate to an uninhabited island off the coast of Queensland for your own safety. Write a list of TWELVE things you’ll be packing to ensure your survival over the next few months on the island.

HINT: Think about how you’ll meet your basic needs: water, food, shelter, defence and communication with the outside world.

Medical resources for emergencies

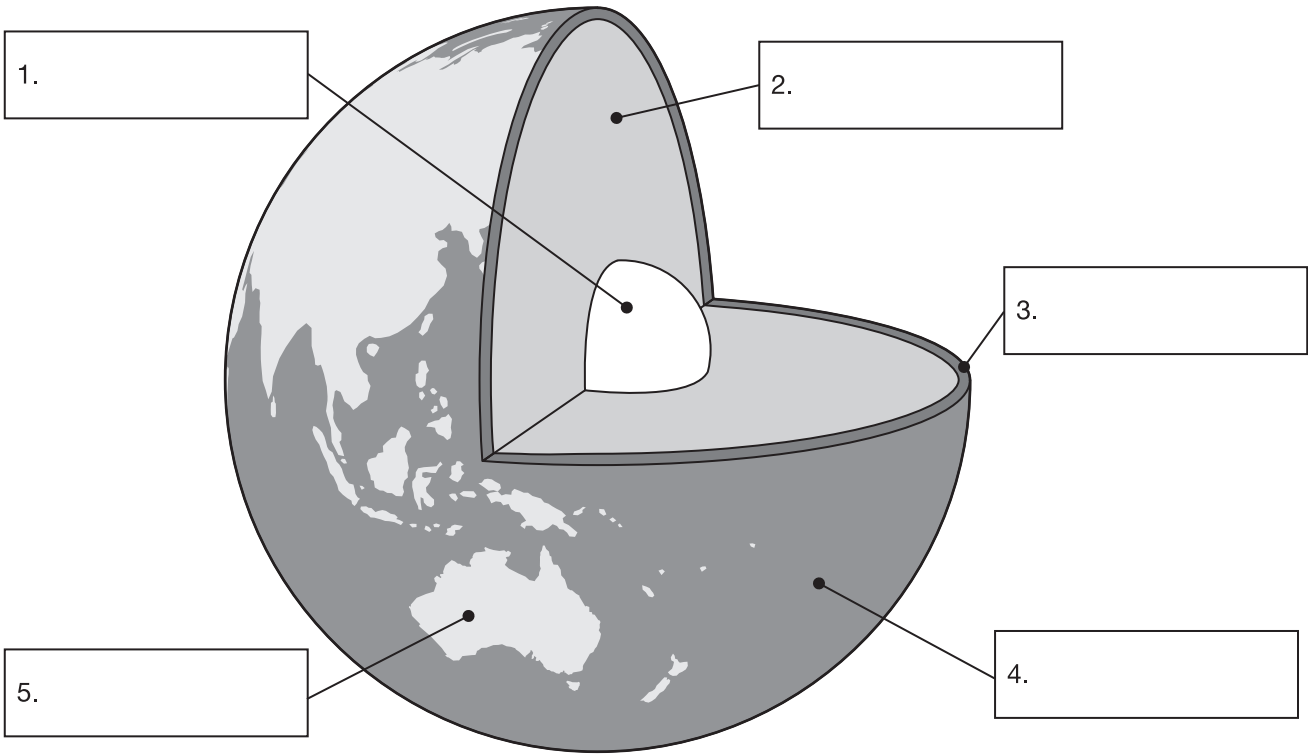
8 List FOUR different illnesses or injuries that could be treated with a first aid kit containing the items in the box.

aspirin	paracetamol	plasters
bandages	sterile syringe	antiseptic
ointment	cotton wool	tape
tweezers	scalpel	vaccines
thermometer	splints	gloves
jellybeans	thermal blanket	sterile water

Cross-section of the Earth

1 Use the features listed in the box to label the diagram of the Earth.

crust continent ocean core mantle



Landforms

2 Using the numbers 1 to 4, arrange these landforms in order of highest to lowest average altitude (or elevation).

_____ hill _____ plain _____ mountain _____ plateau

3 In your notebook, sketch and label a line drawing of each of the landforms listed above, showing their relative shape and height. Use the descriptions in the box to help you.

- landform with steep slopes and jagged points at the skyline
- an elevated land mass with a fairly flat surface
- rounded landform with gentle slopes
- land that is low in elevation and flat or very gently sloping

HINT: You will need to match correctly each landform in Question 2 to its description.

Tectonic terminology

4 Separate this word chain into eight multi-word terms. Write each term below.

supercontinentseismicactivityfaultlinePacificRingofFireRichtersscalecontinentaldriftplatetectonicsseabed

Word functions

5 Underline all of the nouns in the list below. Then circle all of the adjectives. Write each word in your notebook and use a dictionary or glossary to write its geographical definition.

cracks relief collision fissure layered magma massive

surface lithosphere forceful subterranean oceanic tsunami debris

Forces acting on the Earth’s crust

1 Use the terms in the box and the clues in brackets to complete this summary of types of forces.

mud flows	wind erosion	folding	water erosion	ice erosion	earthquakes
rock falls	landslides	avalanche	volcanic activity	faulting	weathering

Atmospheric

- a _____ (wearing away)
- b _____ (erosion caused by water)
- c _____ (erosion caused by wind)
- d _____ (erosion caused by water freezing)

Tectonic

- e _____ (bending to create layers)
- f _____ (cracking)
- g _____ (tremors or shaking)
- h _____ (before or during eruptions)

Gravitational

- i _____ (soil and rock slipping downhill)
- j _____ (stones obeying the law of gravity)
- k _____ (wet dirt running downhill)
- l _____ (ice and snow slipping downhill)

Writing a paragraph

2 Write a paragraph in which you explain the effects of earthquakes on communities. Your paragraph should be 6 to 10 lines long and comprise at least 4 sentences.

HINT: Begin with a topic sentence, then present facts, evidence and ideas to support and further explain the topic. Write a final sentence to strengthen the ideas you have presented.

Transforming a table into a graph

3 In your notebook, present the information from the table in the form of a simple column graph.

Date and place of seismic event	Number of fatalities
1737, Calcutta, India	300,000
1755, Lisbon, Portugal	60,000
1908, Messina, Italy	160,000
1920, Gansu, China	200,000
1988, Armenia	25,000
1990, Iran	60,000
1993, Maharashtra, India	30,000

HINT: Carefully choose a scale for the number of fatalities, which will determine the height of each column. The scale must be divided into equal increments. Label each column with the date and place of the seismic event.

Volcanic terms

1 Match the word beginnings in the boxes to their endings to form a list of volcanic terms.

a General words relating to volcanoes

pyro cra vulcan ext fis dorm act

_____ter _____inct _____ant _____sure _____clastic _____ive _____ology

b Words describing the PRODUCTS of a volcano

m sm ba st la gra a ga pum

_____nite _____eam _____oke _____ice ____sh _____va ____ud _____salt _____ses

c Words describing the ACTIONS of a volcano

ex eru flo bl coll

_____apse _____plode _____ption _____wing _____ast

Lava that has cooled slowly produces igneous rocks with large crystals, such as granite. Rapid cooling produces rocks with softer, smaller crystals, such as pumice.

Classifying rocks

2 Complete these definitions of the three categories of rocks by unscrambling the words in the brackets.

- a Igneous rocks are formed from cooled _____ (gamma) or _____ (vaal).
- b Metamorphic rocks are formed when high _____ (repsruse) and / or temperature causes existing rocks to be _____ (allychemic) altered.
- c Sedimentary rocks are formed from layers of _____ (demsinets) that are under pressure from _____ (sleray) of material above them.

Types of rock

3 Separate this word string into TWELVE types of rock. Then classify each type into its correct category.

gneiss sandstone dolomite basalt granite marble pumice slate limestone chalk obsidian quartzite

IGNEOUS

____ _ i ____

__ a __ a ____

____ _ i ____

____ _ i __ i ____

METAMORPHIC

__ e i ____

__ a ____ _ e

____ _ a __ e

__ u a ____ _ i __ e

SEDIMENTARY

__ o __ o __ i ____

__ a ____ _ _ o ____

__ i __ e ____ _ o __ e

____ _ a ____

Anatomy of a volcano

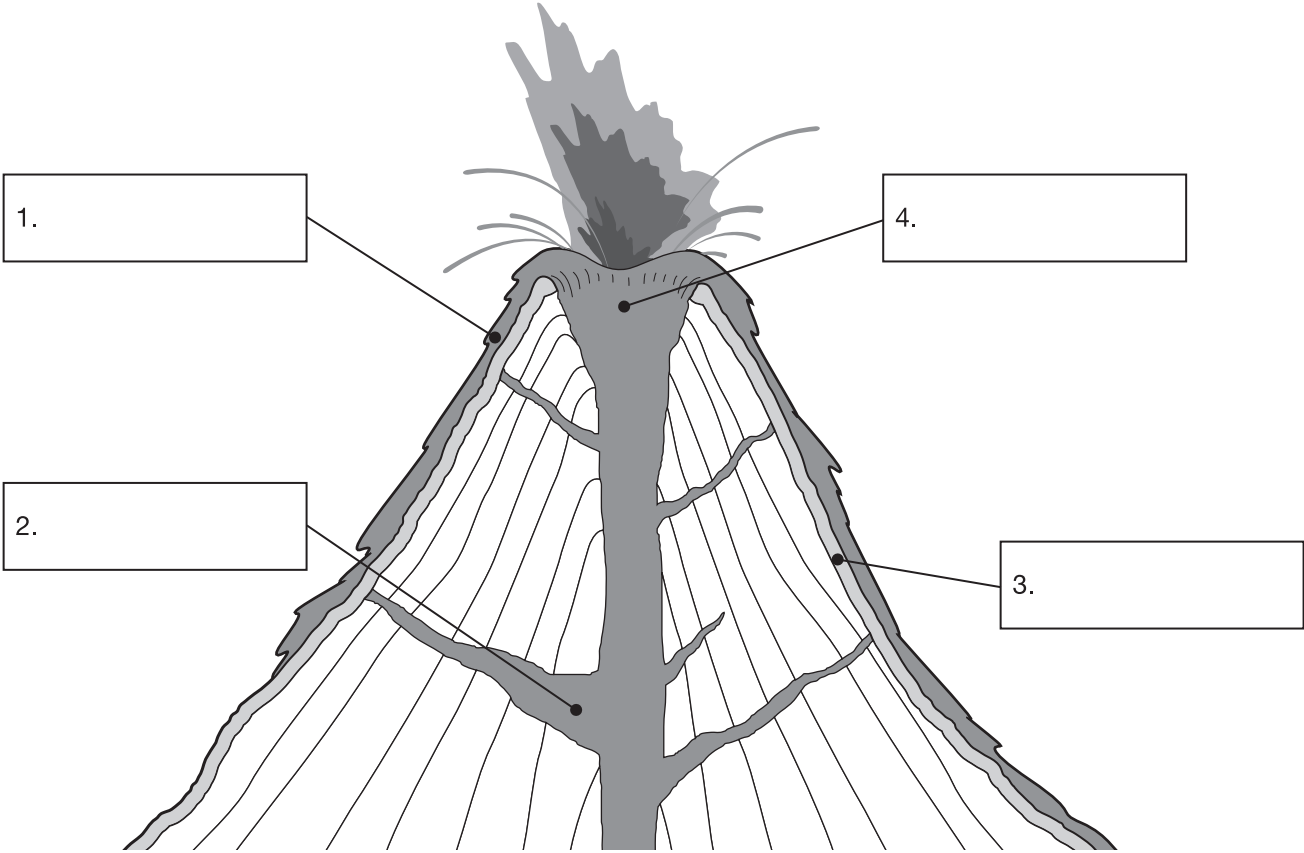
4 Label this diagram of a volcano, using the terms in the box to identify the parts.

vent

magma chamber

lava

ash layer



What's the difference?

5 Explain the difference between the terms in each pair.

- a smoke and steam
- b magma and lava
- c erupt and eject

Where's the volcano?

6 Use an atlas to find the country and hemisphere where each volcano is located, then complete the table.

Volcano	Country	Hemisphere
Mount Etna		
Mount Vesuvius		
Mount St Helens		
Mount Krakatoa		
Mount Fuji		
Mount Vulcan		

The world’s highest mountain

1 The answers to the quiz questions below are encoded using numbers to represent the letters of the alphabet. Fill in the complete number code and write the answers in the spaces provided.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
26	25																							2	1

- a

Name the world’s highest mountain.

14 12 6 13 7

22 5 22 9 22 8 7
- b

In which mountain range is this mountain?

7 19 22

19 18 14 26 15 26 2 26 8
- c

In which country is this mountain range?

13 22 11 26 15
- d

Who was officially the first person to climb this mountain?

22 23 14 6 13 23

19 18 15 15 26 9 2
- e

Which country did this mountain climber call home?

13 22 4

1 22 26 15 26 13 23

Famous mountains

2 Complete these statements about some of the world’s most famous mountains. First, separate the word string into seven names, then use the names to fill in the blanks.

K2MOUNTKILIMANJAROMOUNTMCKINLEYMOUNTCOOKMOUNTFUJIBENNEVISMOUNTACONCAGUA

- a

The world’s second highest mountain is _ _ _ in Pakistan. This mountain claims the lives of one in four people who attempt to reach the summit.
- b

_ _ _ _ _ , Africa’s highest mountain, is located in Tanzania. It stands on its own, and is therefore not part of a mountain range.
- c

_ _ _ _ _ is the tallest peak in North America. It is in Alaska and is home to five giant glaciers.
- d

New Zealand’s highest mountain is _ _ _ _ _ . Its Maori name *Aoraki* means “Cloud Piercer.”
- e

Japan’s tallest mountain, _ _ _ _ _ , is a volcano. It last erupted over 300 years ago.
- f

The Scottish peak _ _ _ _ _ is the highest mountain in the British Isles. It is known by locals as ‘The Ben’ and attracts about 75,000 climbers every year.
- g

The highest mountain in South America is _ _ _ _ _ in Argentina. In 2007, an eleven-year-old American boy named Jordan Romero became the youngest person ever to reach the summit.

Forces that shape mountains

- 3

One of these terms is NOT a tectonic force, but an *effect* of a tectonic force. Circle it.

tsunami faulting volcanic activity folding
- 4

All but ONE of these terms relate to the topic of earthquakes. Circle the odd one out.

seismic vulcanology fault lines Richter Scale
- 5

In your notebook, draw a diagram that shows the difference between the *faulting* and *folding* processes that create mountains.

Two tectonic forces that create mountains are **folding** and **faulting**. Folding occurs because of the pressure created by collisions between continental plates, forcing the Earth’s crust to bend and buckle. Faulting is the cracking of the Earth’s crust, as the edges of the continental plates move, rub together and collide.

Hunt for minerals

- 1 Find all the minerals hidden in the diamond puzzle.
They are hidden forwards, backwards, vertically, horizontally and diagonally.

T I T A N I U M R Z
I J M U I S E N G A M C
K A F N I C K E L Q B T W V
M D L O G L E H J T O U G O E D
S E P N R P A D A E L I R A F P I Y
R I S Y T O M A R D U M T N L R S A L U
E L X V Q A M I L N W N V I A D T E Z P
P V H I R O N A Z X E I N M U R K Z A C
P E J B E K R N I B E T E H P F G N P N
O R L V T E H U G D M T N W B Q L M R I
C E I E M L J T B W H R A E C I S E N Z
E T F E N D R A C Y F L R G M Z T W E S
F J O X S O E O S Z J Y N E A S O R I L
E N D F W T T M P L U S O A B I P A
S R I K G S C W E T P B Y H S P
I O R L H D X O R A N P E O
O L E M U N I L A P P I
U A P E B A L A X L
Q P B O R S N U
R I Q J H Z
U S X A
T L

- DIAMOND
- SILVER
- TIN
- AMETHYST
- TOPAZ
- ONYX
- IRON
- AGATE
- QUARTZ
- SANDSTONE
- MARBLE
- LIMESTONE
- ZINC
- EMERALD
- SAPPHIRE
- RUBY
- NICKEL
- GRANITE
- JADE
- JASPER
- GOLD
- MAGNESIUM
- COPPER
- GARNET
- LEAD
- PERIDOT
- FLINT
- ALABASTER
- BERYL
- OPAL
- TURQUOISE
- LAPIS LAZULI
- TITANIUM

Classifying minerals

- 2 Now that you have found all the minerals, classify them into the tables below as either metals or stones.

Metals

HINT: In the puzzle, the stones were hidden diagonally.

To be classified as a stone, the substance must have a crystalline structure.

Stones		

The world’s oceans and seas

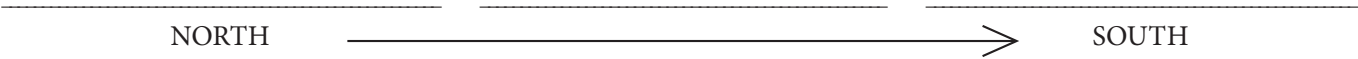
- 1 Name the three major oceans of the world.
- _____
- 2 Complete the names of these famous seas by adding the correct vowels.
- MDTRRNN _____ BLCK _____
- NRTH _____ DD _____

Australian oceans and seas

- 3 Write these Australian oceans and seas in the correct order moving from North to South. Use an atlas to help you.
- Southern Ocean

Tasman Sea

Coral Sea



Features of oceans

- 4 Circle the terms that DO NOT describe features of various oceans around the world.
- currents

canopy

floor

salt water

ozone

tides

icebergs

Marine words

- 5 The word *marine* means ‘of the ocean’. Complete these words beginning with *mar*, using the clues and the synonyms (shown in brackets).
- mar _ _ _ _ a person who frequently voyages on the sea (sailor)
- mar _ _ _ _ a place set aside as a storage and launching facility for boats (wharf)

Sea-going vessels

- 6 Write the vessels in each set in order from largest to smallest.
- a dinghy, fishing trawler, frigate

b cargo ship, lifeboat, tugboat

c canoe, passenger liner, patrol boat

d houseboat, destroyer, ski boat

- ## Understanding knots
- 7 Insert the words below into the spaces provided to complete the passage.

mile

convert

measurement

1.15

multiply

A knot is a unit of _____ for a ship’s travelling speed. It equals one nautical _____ per hour.

A nautical mile is equal to _____ land miles. To _____ miles per hour into kilometres per hour, _____ the figure by 1.609344.

8 Use the information above to calculate the distance in kilometres covered by a ship in one hour if it is travelling at a speed of 15 knots. Round off your answer correct to two decimal places. _____
- 26

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Human use of oceans

9 Fill in the names of the ocean-related activities using the clues provided.

- a putting holes in the ocean floor and pumping out ‘black gold’

o _ _ d _ _ _ _ _ _ _
- b harpooning large marine mammals

w _ _ _ _ _ _ _
- c catching too many edible, scaly marine creatures

o _ _ _ - f _ _ _ _ _
- d setting off radioactive bombs underwater

n _ _ _ _ _ _ _ t _ _ _ _ _
- e permission for traditional owners of ocean areas to fish

I _ _ _ _ _ _ _ _ _ r _ _ _ _ _

10 Complete the table by correctly grouping these terms.

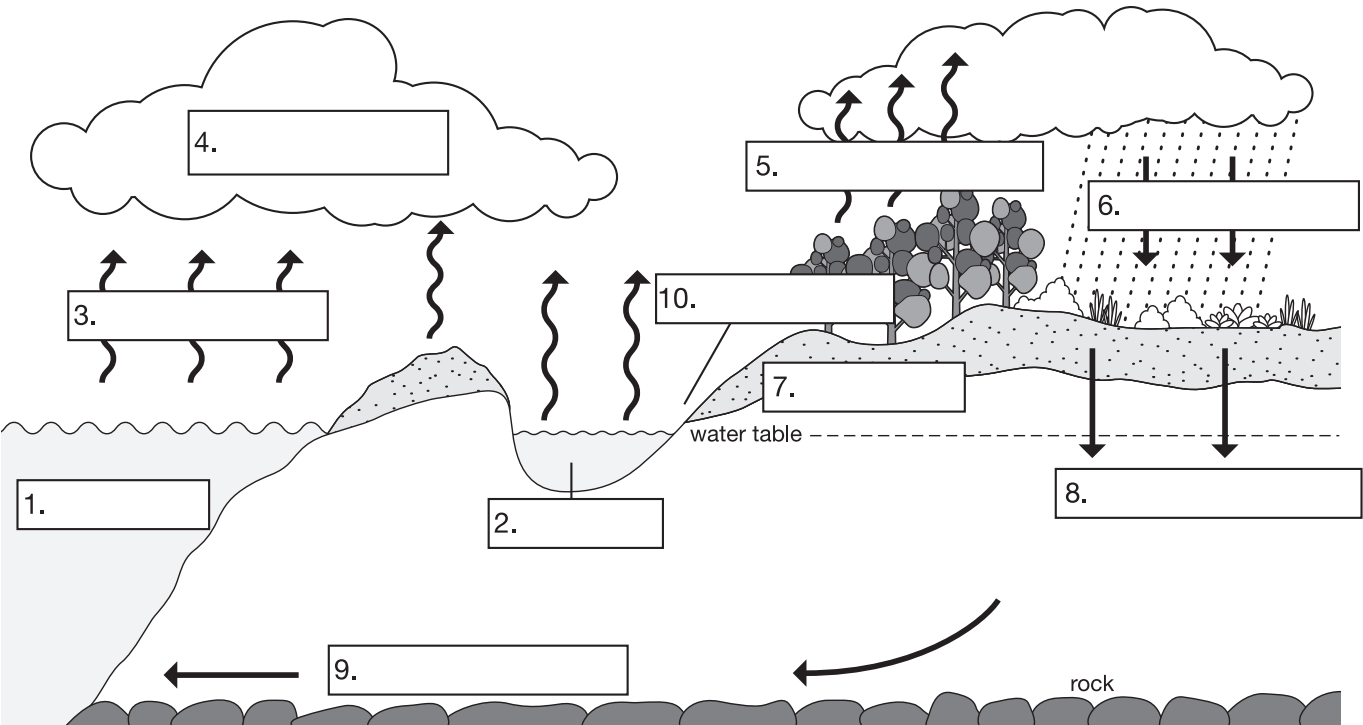
driftnets	refugees	water sports	exports	natural gas reserves
border protection	merchant shipping	snorkelling	oil platforms	long lines
cargo ships	minerals	whale watching	illegal entry	catch limits

Fishing	Mining	Tourism	Trade	Immigration

The water cycle

11 From your own knowledge, label this diagram illustrating the ocean’s role in the water cycle. Use the terms in the box as labels.

ocean	land	lake	infiltration	evaporation
run-off	condensation	precipitation	transpiration	groundwater



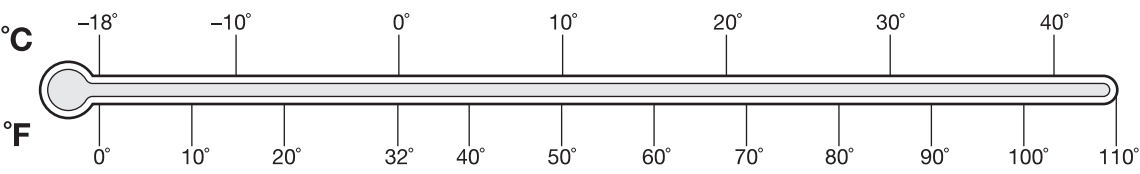
Geographical terms

- 1 Rewrite each term with its correct spelling, using the pronunciation clues given. Then, place each term before, between or after the additional words in the brackets to make phrases used in geography. The first one has been done for you.
- | | | | | |
|---|-----------------------|-------|-------------------|---------------|
| a | soul-lar | solar | (powered) | solar powered |
| b | ray-dee-ant | | (heat) | |
| c | new-clee-ar | | (anti-, protests) | |
| d | poll-oo-shon | | (water) | |
| e | at-moss-fear | | (Earth's, the) | |
| f | en-urge-ee | | (efficiency) | |
| g | teck-noll-ogee | | (information) | |
| h | pro-sess-ez | | (natural) | |
| i | cly-mett | | (change) | |
| j | see-suns | | (four, the) | |
| k | carr-ben die-ock-side | | (emissions) | |
| l | foss-ell few-els | | (conserving) | |

Solar and atmospheric processes

- 2 Group these sets of word fragments together into two-word terms relating to solar and atmospheric processes.
- | | | | | | | |
|---|------|--------|-------|-------|-----|--|
| a | ect | green | eff | house | | |
| b | mate | change | cli | | | |
| c | war | bal | glo | ming | | |
| d | ergy | sol | en | ar | | |
| e | vio | ult | light | ra | let | |

Measuring temperature



- 3 Use the diagram to estimate the following Fahrenheit temperatures in degrees Celsius.
- | | | | | | |
|---|------|---|------|---|-------|
| a | 10°F | b | 60°F | c | 100°F |
|---|------|---|------|---|-------|
- 4 Now convert the following Celsius temperatures to degrees Fahrenheit.
- | | | | | | |
|---|-----|---|------|---|------|
| a | 0°C | b | 36°C | c | 16°C |
|---|-----|---|------|---|------|

The Earth’s ozone layer

1 Complete the missing words in the passage, using the first letters to guide you.

Ozone is a g_____ that acts as a protective shield in our atmosphere. The ozone layer filters the sun’s r_____, preventing about 99% of the harmful u_____ light from reaching the Earth’s surface. The screening power of the ozone is becoming l_____ effective due to its destruction by polluting gases. These human-made gases are called ‘chlorofluorocarbons’ or CFCs. The d_____ of the ozone means that humans and wildlife species are exposed to radiation that can now penetrate the ozone l_____. UV rays can cause damage to the chromosomes of living things, and are responsible for the formation of certain d_____, such as skin cancer. One area particularly at risk is Antarctica, which lies directly under the main h_____ in the ozone layer. The UV light is causing damage to microscopic organisms called ‘phytoplankton’. Phytoplankton use sunlight to produce food to sustain themselves and grow. They are eaten by tiny m_____ animals called ‘krill’ and are their only food source. Many marine species rely on krill, which is the most important food source in the Antarctic f_____ web.

Defining terms

2 Using the information in the passage above and a dictionary, write sentences to define each of these terms.

phytoplankton _____

ultraviolet light _____

polluting _____

CFCs _____

Sources of CFCs

3 Use these words to create three two-word terms that are sources of CFCs.

plastics spray conditioners air cans foam

a _____

b _____

c _____

Food chain

4 Draw a simple food chain that includes: whale, phytoplankton, sunlight, krill.

Food chain:

What can we do?

5 Suggest THREE practical actions someone your age could take to help minimise further damage to the ozone layer.

Classifying clouds

- 1 Clouds are classified as either layered or convective. The category names are *stratus* and *cumulus*. Which is which?
- a _____ clouds are named from a Latin word meaning ‘layer’.
- b _____ clouds take their name from a Latin word meaning ‘piled up’.

Clouds are divided into four main groups, according to the altitude at which they are found. Storm clouds that stretch from low to high altitudes in one gigantic mass are called *cumulonimbus* clouds – a word formed from two cloud types.

- 2 Break the letter chain into the four cloud categories. Write the categories below.

cirruscumulusnimbusstratus

Condensation and precipitation

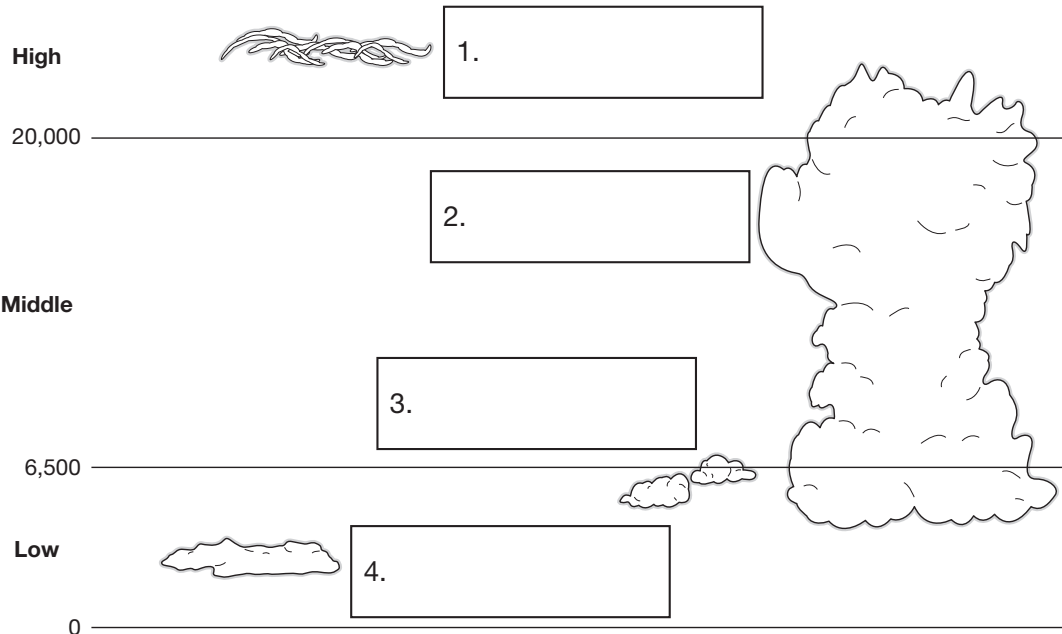
- 3 Complete the following sentences.
- a The changing of water vapour from a gas to a l _ _ _ _ _ is called condensation.
- b Condensation and precipitation are part of the water c _ _ _ _ _.
- c During this process, moisture circulates around the h _ _ _ _ _ , atmosphere, lithosphere and biosphere.
- 4 Choose the correct word from the box to complete the explanation of condensation and precipitation.

precipitation supported vapour droplets increase clouds suspended heavy surface

Clouds are made up of water _____, which consists of _____ that are so small that they can remain _____ in the air. When _____ hang low over the Earth’s _____, they are recognised by us as fog. For _____ to occur, the water droplets that form clouds have to _____ their size. Gradually, these droplets become so _____ that they can no longer be _____ by the air.

Name that cloud

- 5 Label the different cloud types in this diagram.



Cloud seeding is an experimental process in which meteorologists put various elements into the air to encourage condensation to occur. Cloud seeding is designed to provide particles around which water droplets can form, therefore leading to increased rainfall.

Drawing cloud combinations

6 Draw one type of cloud from each of these three categories.

High clouds

Cirrocumulus
Cirrostratus

Middle clouds

Altostratus
Alto cumulus

Low clouds

Nimbostratus
Stratocumulus

A cloud’s altitude is measured from the ground to the height of its base.

Storm clouds

7 Rewrite these sentences, switching the order of the words in each pair shown in bold.

a When **air cold** pushes under a mass of **air warm**, the lighter warm **rises air** and **occurs condensation**.

b This process of **movement air** creates **clouds cumulonimbus** in the atmosphere.

c These large, **clouds dense** gather to **thunderstorms create**, with **rainfall heavy** and sometimes hail.

Describing and defining

8 Write your answers to these questions in full sentences, using the words provided in any order you choose.

a What is a cloud?

mass droplets floating atmosphere visible

b Why do cumulus clouds appear to be white?

sunlight droplets tiny crystals reflect

c How does hail form?

water vapour condenses freezes rapidly falls

Features of desert environments

- 1 Circle the correct synonym for the words in **bold** in these sentences about desert environments.
- a The main feature of all deserts is their **aridity**.
 - b Deserts have very low annual **precipitation**.
 - c Wind can wear away rocks by the process of **abrasion**.
 - d Sand dunes are formed by the **deposition** of fine particles.
 - e **Weathering** gradually turns large rocks into smaller rocks.
 - f There is little **vegetation** in deserts due to the lack of water.
 - g The **absence** of clouds leads to high daily temperatures.

HINT: A synonym is a word that has the same or similar meaning.

heat	dryness	flatness
rainfall	temperature	hydration
traction	coercion	friction
dumping	motion	action
rainfall	sunlight	wearing away
vegetables	flora	growth
abundance	altitude	lack

Managing deserts

- 2 Write a sentence explaining how we can try to fix each of these problems in desert environments.

Damage to sand dunes from tourism: _____

Wind erosion of sand dunes: _____

Lack of access to fresh water: _____

Finding water in the desert

Over 40% of the Australian continent is classified as desert.

- 3 Suggest ways in which a desperately thirsty person could find drinkable water in the desert, using the clues.

a

dig sand

b

roots tree

c

condensation collect night

d

birds animals follow

Deserts of the world

4 Match these deserts with their locations by writing the correct letters in the spaces.

- a

Simpson

USA
- b

Gobi

Australia
- c

Sahara

Mexico
- d

Chihuahuan

Australia
- e

Kalahari

Argentina
- f

Syrian

North-western Africa
- g

Gibson

China and Mongolia
- h

Colorado

South-western Africa
- i

Patagonian

Saudi Arabia, Jordan, Syria and Iraq

HINT: If you're not sure about some of them, do the ones you know first and work out the others using the process of elimination.

An environment is classified as a desert because of the amount of annual rainfall it receives. There are hot deserts and cold deserts. The Maudlandia Tundra region in the Antarctic is a desert, and is one of the coldest places on Earth.

Drawing a desert

5 Choose ONE of the deserts listed above and sketch a landscape showing some of that desert's features and landforms.

HINT: You will need to research images of these deserts – they are not all sand!

Shades of meaning

6 Explain the differences in meaning between the terms in each pair.

- a

windward and leeward

- b

deposition and deflation

- c

weathering and erosion

Grasslands facts

- 1 Rearrange these sentence fragments to write complete sentences about grasslands. Remember to use correct punctuation.
- a savanna one type of tropical grassland is the
- b prairies of North America are known as the grasslands
- c wettest tropical summer season is the in savannas
- d American prairies bison buffalo on the and deer once grazed
- e often occur at the end of grass fires winter
- f are the biggest lions of the grasslands carnivores African
- g provide a herbivores grasslands good home to

Types of grasslands

- 2 Circle the most appropriate word from each pair to complete these statements about grasslands.
- a The two main types of grasslands are tropical savannas and **temperate** / **temperature** grasslands.
- b Tropical grasslands get very little **rains** / **rainfall** throughout the year.
- c Tropical grasslands are located between the **Tropical** / **Tropics** of Cancer and Capricorn.
- d The Great Steppe is a **treeless** / **treed** grassland in Russia.
- e The Veldt is a grassland in the southern part of the African **continent** / **country**.
- f Grasslands have **relative** / **relatively** few trees and shrubs compared to mountains.
- g Many varieties of mammals live **over** / **on** grasslands.

Locations of grasslands

- 3 Unscramble the vowels and consonants below to spell the names of three CONTINENTS that contain grasslands.
- a oaeia nrthmrc _____ (two words)
- b aia frc _____ (one word)
- c auaia strl _____ (one word)

*The word **savanna** is sometimes spelled **savannah** – both spellings are acceptable.*

Inhabitants of grasslands

- 4 Each of these groups of animals belongs to one of the grassland habitats identified in question 3. Write the correct continent for each group of animals.
- a elephant, antelope, cheetah, giraffe, hippopotamus, hyena, leopard, ostrich, vulture, wildebeest, zebra
- _____
- b dingo, emu, kangaroo, wallaby, wombat, camel, horse, rabbit
- _____
- c badger, bison, jack rabbit, bumble bee, buffalo, condor, coyote, deer, eagle, elk, wolf, squirrel, skunk
- _____

More grasslands facts

- 5 Rearrange these sentence fragments to form complete sentences about grasslands. There are two sentences to be made from each set of words. The beginnings have been done to help you.
- a kill and and scavengers eat herbivores predators carnivores are
- Carnivores _____.
- Two types of _____.
- b have now been South Africa into is a grassland farmland Park
- grasslands made in National
- Many _____.
- Kruger _____.
- c the main deserts desertification of grasslands threatening
- is overgrazing grasslands problems is called turning into
- The process _____.
- One of _____.
- d are needed a decrease to help have caused for future generations
- and poaching in the numbers of some species animals protect the grassland
- Hunting _____.
- Stricter laws _____.

Other names for grasslands

- 6 Break the word chain into seven separate plural terms for grasslands. Write the terms below.

savannaspampasplainssteppesprairiesveldtscampos

Different types of wetlands

1 Wetlands are valuable water-cleansing areas where the water is only just above sea level and covers the ground for much of the year. Write the correct type of wetland listed in the box next to each definition.

mudflats and saltmarshes	bogs and fens	estuaries and tidal flats	swamps and marshes
--------------------------	---------------	---------------------------	--------------------

- a

cool climate wetlands, often containing peat
- b

warm climate wetlands, containing small trees and grass
- c

wetlands located near the sea
- d

muddy wetlands near lakes and rivers

Wetlands word puzzles

2 Solve these word puzzles using the letter shapes and clues provided. The first one has been done for you.

- a

wetlands

Areas of low-lying land covered with water for most or part of the year.
- b

The place where a particular plant or animal species naturally lives and thrives (fifth letter is a **t**).
- c

Trees that thrive in salt water environments (first letter is an **m**).
- d

A well-known alligator-infested swampland in Florida, USA (starting with a capital **E**).
- e

A national park wetlands area in Australia’s Northern Territory (first letter is a capital **K**).

The water cycle of the wetlands

3 Complete this diagram of the wetlands water cycle by adding arrows to indicate the direction in which the water moves through the cycle. Add the following labels: evaporation, condensation, transpiration, precipitation, run-off.



Inhabitants of wetlands

4 Circle 15 living things that can be found in wetland areas.

ducks	fish	bears	frogs	orang-utans	elephants	bacteria
water rats	giraffes	alligators	camels	shellfish	sharks	mosquitoes
emus	polar bears	otters	dogs	platypuses	swans	snakes
cattle	lions	beavers	herons	geese	cats	monkeys

Spelling rainforest words

1 Circle the correct spelling in each of these sets of words.

- a tropical

tropacal

tropacil
- b canapy

canipy

canopy
- c ecosistem

ecasystem

ecosystem
- d habatat

habitat

habetat
- e photosynthesis

photosinthesis

photosinthesys
- f speshies

species

spechies
- g pollination

pollanation

pollenation
- h moss

mos

mosse
- i fernes

ferns

furns
- j humiditty

humidity

humitivity

Understanding photosynthesis

2 Explain the process of photosynthesis.

HINT: photo = light;
synthesis = to make something.

A three-storey forest

3 Draw a line through the INCORRECT descriptors of each storey (or level) of a rainforest. There are seven incorrect statements in total.

GROUND LEVEL

- is very sunny and warm
- encourages the growth of moss
- encourages the growth of ferns
- contains lichens
- contains fungi
- is a good place for photosynthesis
- is called the forest floor

LOWER STOREY

- has many green plants
- has few green plants
- has lots of light available
- is dark and wet
- is dry and dusty
- has humid air
- is the middle layer of the forest

UPPER STOREY

- covers the forest like an umbrella
- is located near the ground
- may be woven together with vines
- keeps the sunlight from areas below
- is full of moss and lichens
- is called the canopy
- is the ideal place for photosynthesis

Rainforest facts

4 Complete these sentences about rainforests, using the definitions of the missing words to help you.

- a

Tropical rainforests are located in countries near the E _ _ _ _ _ . (imaginary line encircling the Earth)
- b

Bright flowers and strong scents attract insects to help rainforest plants p _ _ _ _ _ . (transfer pollen)
- c

Rainforest plants have waxy leaves to help them keep off any excess w _ _ _ _ . (H₂O)
- d

Many of the world’s rainforests are vanishing due to d _ _ _ _ _ . (the cutting down and removal of trees)

Finding rainforests



5 Write the letter **R** or draw a fern leaf next to each country in which rainforests grow.

- Saudi Arabia

Fiji

Australia

Greenland
- Canada

Congo

Indonesia

Sweden
- Ecuador

Japan

Egypt

Venezuela

Primary forest (or ‘virgin forest’) is the term we use to describe a rainforest in its natural condition.
Secondary forest is a forest that has been cleared of some or all of its trees in the past, but has regrown.

Definitions of mountain terms

- 1 Match each term with its definition by writing the correct numbers next to the terms in the first column.
- | | |
|-------------------------|---|
| _____ glacier | 1 the gradual breakdown of rock into smaller particles by various processes over time |
| _____ avalanche | 2 a rounded mountain that doesn't break through the Earth's surface |
| _____ alpine | 3 the centre of the Earth, composed of a solid inner part and a molten outer layer |
| _____ altitude | 4 the layer of the Earth between the crust and the core |
| _____ snowline | 5 the sudden rapid movement of ice and snow downhill |
| _____ range | 6 superhot liquid rock below the Earth's crust |
| _____ erosion | 7 a chain of islands formed from the peaks of a mountain range submerged in the sea |
| _____ spur | 8 the area above a mountain's tree line |
| _____ weathering | 9 height above sea-level |
| _____ rift valley | 10 a mass of ice and compacted snow that moves very slowly down a slope over time |
| _____ subduction | 11 the picking up and removing of weathered rock material |
| _____ tectonic activity | 12 an undersea mountain |
| _____ ridge | 13 a chain of mountains |
| _____ plates | 14 the rate at which the air temperature gets lower when altitude increases |
| _____ seamount | 15 flowing magma that has reached the Earth's surface |
| _____ archipelago | 16 a flat area of land that is of a uniform height above sea level |
| _____ mantle | 17 a sharp peak on a glacial mountain |
| _____ plateau | 18 a ridge of land higher than its surroundings |
| _____ fault | 19 the harsh rubbing action of rocks, ice and other hard materials on the Earth's surface |
| _____ magma | 20 the solid, outer layer of the Earth, made up of large segments called 'plates' |
| _____ crust | 21 a valley formed from the collapsing of land between two faults that lie side by side |
| _____ fold | 22 the movement, breaking, folding and colliding of the Earth's crustal plates |
| _____ dome | 23 high land forming a line, occurring between two valleys |
| _____ abrasion | 24 the massive sections of the Earth's crust |
| _____ lava | 25 the division between snow-covered and uncovered ground |
| _____ core | 26 description of the state of an inactive volcano |
| _____ sediment | 27 a crumple in the Earth's crust from pressure and movement under continental plates |
| _____ dormant | 28 eroded material that has been deposited in water |
| _____ lapse | 29 the movement of one crustal plate beneath another |
| _____ horn | 30 a torn place in the Earth's crust that is broken or unstable |

The formation of mountains

- 2 Reorder these sentence parts so each statement about the formation of mountains makes sense. Indicate the correct order by writing 1, 2 and 3 next to each phrase in the spaces provided.
- | | | |
|---|--------------------------------------|--|
| a which make up the Earth's crust _____ | the movement of the plates _____ | mountains have formed because of _____ |
| b by the action of _____ | mountains can be shaped _____ | wind, water and ice _____ |
| c a mountain range _____ | occur in long chains to create _____ | a group of mountains can _____ |
| d the inner core of the Earth _____ | while the outer core is molten _____ | is made of solid metal _____ |
| e the slow movement of _____ | creates convection currents _____ | hot material from the core _____ |
| f the plates of the Earth's crust _____ | convection currents can cause _____ | to move and collide _____ |

Mountain puzzle

3 Circle the following words in the puzzle. They are hidden forwards, backwards, vertically, horizontally and diagonally.

glacier	avalanche	altitude	snowline	range	erosion	spur	weathering
rift valley	subduction	tectonic activity	ridge	plates	seamount	archipelago	mantle
plateau	fault	magma	crust	fold	dome	core	abrasion
lava	sediment	dormant	lapse	horn	earthquake	water	ice
hill	tree	plain	ice-cap	valley	slopes	tropical	snow
forest	land	frost	lake	peak	alpine		

*
 H
 S * I
 P K A E P
 L A L P I N E
 A V S * * E M
 * R A N G E R * A E
 S U L O N R O H L K
 U P A W * * S T W A
 G A S S N O W L I N E U
 U C L * * C * * O A A Q
 A * * A * H I L L N M T H
 T * C * B I C E C A P F R H T
 N S * R P L A I N * * O O E R
 E A A S * Y * * E * * R D R A
 L M S L U T * E * * R * E * I E
 T I T T B R * S L O P E S * N A
 O D S I D O M E * L * * T T G Y *
 L * * * S Y N A E O T U P U A E T A L P L L A *
 A R I D G E * * S R U C I R E T A W V * U A S *
 R I C E E L A V A F D T C * L A P S E T A N T A
 U * * R P L A T E S E I A * T * * D L O F D S N
 * * T S E A M O U N T O L * N * A M G A M I U D
 S L A K E V T E C T O N I C A C T I V I T Y R E
 E A R C H I P E L A G O I K M C O R * E R O C S

Mountain ranges

4 Starting from the peak and working clockwise, use the leftover letters to spell the names of seven mountain ranges.

Mount Kosciuszko was named in 1840 by Count Strzelecki, an explorer and geologist. The name was chosen in honour of a Polish freedom fighter. Prior to 1840, the mountain was named 'Tar Gan Gil' by the Aboriginal people of the region.

The air contains less oxygen at higher altitudes. The Sherpa people of the Himalayas have adapted to the thinner atmospheric conditions by developing greater lung capacities than people who live in places closer to sea level.

Questions about coral reefs

- 1 Reorganise the following groups of words to form questions, then rewrite each one with correct punctuation. Use each answer to help you work out the question.

a small animals are forming coral reefs what responsible for

Answer: **polyps**

b these tiny what do animals eat

Answer: **algae**

c coral made which substance is from

Answer: **limestone**

d things do help coral what living reefs support

Answer: **fish**

e reefs grow do quickly or coral slowly

Answer: **slowly**

f are the world's reefs in which ocean most of coral

Answer: **Pacific**

g threaten coral what is human activity that one could a reef

Answer: **tourism**

Animals of the reef

- 2 Circle **TRUE** or **FALSE** for each statement.

a The crown-of-thorns starfish can grow as many as 21 arms.

TRUE **FALSE**

b A common type of reef shark is the Great White Shark.

TRUE **FALSE**

c Dolphins breathe underwater.

TRUE **FALSE**

d Most baby turtles don't survive to adulthood.

TRUE **FALSE**

Reef life

- 3 Complete the following passage by circling the most geographically and grammatically correct word or phrase from each set of terms.

A coral reef has great biodiversity, supporting a **mixture / range / whole lot** of life forms. Reefs are home to many different **kinds / species / types** of fish. These fish graze on the weeds and grasses that **are / grow / sit** on the coral. It is important that reefs **keep / maintain** a rich supply of food sources for sea creatures that **want / depend on / need** them for their **life / survival**. Sharks and other **killers / predators / hunters** are at the top of the food chain. They eat **big / large / huge** fish, which in turn eat **tinier / littler / smaller** fish. Coral reefs also **provide / give** a home to sea turtles and dolphins. Other **mammals / creatures / animals** that live near reefs are dugongs and whales. If the reef **ecosystem / area / system** is **damaged / wrecked**, the lives of thousands of **water / marine / oceanic** creatures will become **difficult / endangered**.

The organisms that live on coral reefs can be described as 'interdependent', which means that they all depend on each other for their survival.

Coral reef word pairs

4 Use one word from each box to form word pairs that match the definitions below.

- marine
- box
- sea
- calcium
- symbiotic
- limestone
- food
- primitive
- polyp
- growth
- fragile

- carbonate
- rate
- ecosystem
- life
- organism
- relationships
- colony
- anemone
- chain
- jellyfish
- skeleton

There is no known antidote for the venom of the blue-ringed octopus, which is deadly to humans.

- a Another name for limestone

b A system where both partners benefit equally

c A delicate environment

d A reef creature that has a poison deadly to humans

e A simple type of living thing

f A community of tiny animals that make coral

g A plant-like animal of the reef

h How fast or slow something develops

i A way of illustrating which animals eat each other

j A term meaning all the living organisms in the sea

k The rigid part of a polyp that remains after its death
- c _____

s _____

f _____

b _____

p _____

p _____

s _____

g _____

f _____

m _____

l _____
- c _____

r _____

e _____

j _____

o _____

c _____

a _____

r _____

c _____

l _____

s _____

Whose food?

5 Match the sea creatures with the correct foods by choosing from the box to complete the sentences.

- plankton large fish algae and coral polyps small fish

- a Small fish eat _____
- b Predator fish eat _____
- c Whales eat _____
- d Large fish eat _____

Who's eating whom?

6 Reorganise these organisms to create food chains. Use the point of each arrow to indicate the eaters, as shown in this example: cheese → mouse → cat

- a sharks, small fish, algae, large fish
- _____ → _____ → _____ → _____
- b crabs, humans, tiny fish, seaweed
- _____ → _____ → _____ → _____

Hidden homes and favourite foods

1 Unscramble the letters to identify which place each of these endangered species calls home. Then, use the food lists in the box below to complete the final column. The number clues tell you how many letters are in each food listed. The first one has been done for you.

Endangered species		Home	No. of letters in food list	Food list
a	orang-utan	(donesinai)	Indonesia	(6,5,4,4,4) J
b	panda	(hanic)		(6,5,4,4,4)
c	tiger	(umastra)		(4,4,7,7,6)
d	emperor penguin	(catnitcara)		(4,5,5,6)
e	gorilla	(goonc)		(5,4,6,4,7,4)
f	fur seal	(sitaralua)		(5,7,9,4)
g	grizzly bear	(thorn eriamac)		(5,5,4,5,7)
h	reindeer	(palland)		(6,5,5,6)
i	elephant	(yenka)		(5,6,5,4,5,8)
j	lemur	(gadmascara)		(6,7,5,7)

List A	leaves, flowers, fruit, insects	List F	bamboo, honey, eggs, fish, yams
List B	roots, fungi, fish, honey, mammals	List G	deer, pigs, badgers, rabbits, cattle
List C	fish, krill, squid, shrimp	List H	grass, leaves, twigs, bark, fruit, seedpods
List D	squid, octopus, shellfish, fish	List I	fruit, nuts, leaves, bark, insects, eggs
List E	lichen, fungi, grass, shrubs	List J	roots, fruit, shoots, leaves, termites, bark

Names of endangered species

2 Fill in the missing letters to complete these names of some other endangered species. Then circle the one that is found in Australia.

a	p _ _ _ _ bear	b	wh _ _ _ r _ _ _ _ _ _ _ s
c	b _ _ _ w _ _ _ _	d	_ _ ee k _ _ _ _ _ oo

Endangered plants

3 There are many endangered plant species as well as animal species. Why is it important that we preserve endangered plants? Write a paragraph in response to this question, using words and phrases from the box mixed with your own words.

food for animals habitats food webs food for humans valuable medicines beauty

Understanding terminology

- 1 Explain the meaning of each **bold** term in the following sentences by examining the context. Give examples where appropriate.
- a Bushfires are more likely to start through **natural processes** when the **vegetation** is dry and easily **ignited**.
 - b The worst fire-fighting **conditions** are hot days with strong, **unpredictable** winds.
 - c People are often **evacuated** from **residential areas** when bushfires are burning nearby.
 - d Fires also affect **fauna** by destroying their **habitats** and preventing them from reaching fresh water.
 - e **Arson**, the cause of many **devastating** bushfires, is a serious criminal offence.
 - f **Back-burning** is practised in certain areas to use up **potential fuel** for bushfires.
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

Order of events

- 2 What happens first? Organise these pairs of terms in their correct logical order by writing 1 or 2 in the spaces.
- | | | | |
|--------------------------|---------------------|----------------------|------------------|
| a _____ arson | _____ fire fighting | b _____ fire | _____ evacuation |
| c _____ lightning strike | _____ scrub fire | d _____ waterbombing | _____ igniting |

Negative effects of bushfires

- 3 Rewrite these sentences, correcting any errors in expression, grammar, spelling and punctuation.
- a fires produce heaps of haze and smoke and heat and ash
 - b they pressent a danger to fire fiters and can result in injurys and loss of lifes
 - c fire fighting costs alot of time, manpower and resources
 - d bushfires sometimes force people to leeve they're homes
- _____
- _____
- _____
- _____

Functions of city precincts

1 Write the features listed in the box next to the correct precincts.

police headquarters	shops	cathedral	factories	government offices	banks
stock exchange	flats	university	amusement parks	insurance offices	docks
houses	parks	apartments	fisherman’s wharf	shopping mall	zoo
mosques	colleges	schools	sewage plant	social services offices	gasworks
courthouse	markets	cinema	marina	churches	power grid

PRECINCTS	FEATURES			
industrial	_____	_____	_____	_____
recreational	_____	_____	_____	_____
commercial	_____	_____	_____	_____
financial	_____	_____	_____	_____
residential	_____	_____	_____	_____
educational	_____	_____	_____	_____
religious	_____	_____	_____	_____
waterfront	_____	_____	_____	_____
judicial	_____	_____	_____	_____
administrative	_____	_____	_____	_____

Reading a city map

2 Look at the map and answer these questions.

- a

Is the Royal Exhibition Building north or south of Chinatown?

- b

Which avenue runs closest to the Melbourne Cricket Ground?

- c

How many bridges across the Yarra River are shown on this map?

- d

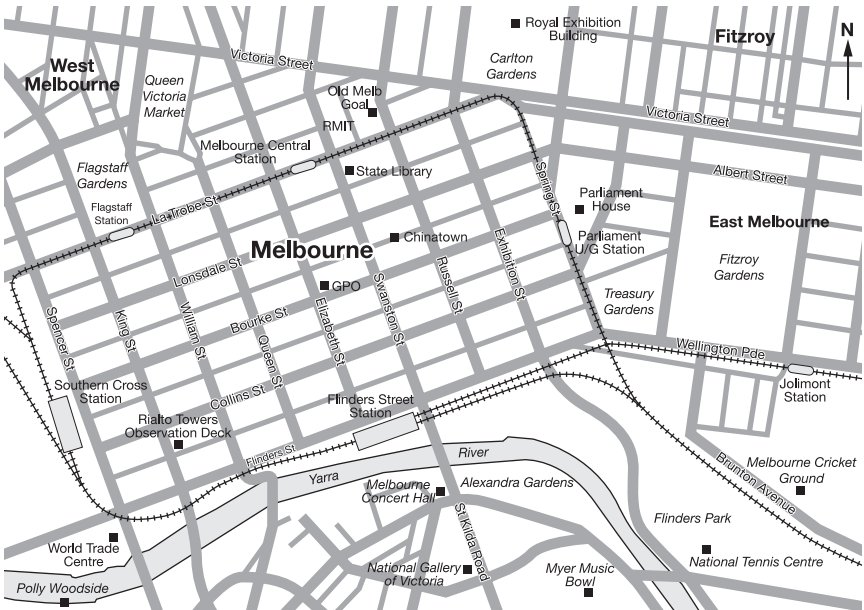
Which two streets are named after famous British queens?

- e

List THREE sports, arts or cultural facilities located south of Wellington Parade and east of St Kilda Road.

- f

Which politically important building in East Melbourne has its own underground railway station?



Designing a city

3 In your notebook, create a design for a coastal city’s transport infrastructure (roads, railways, monorail, airports and docks). Make an A4 sized drawing and mark on it logical locations for each of the precincts listed in the first activity.

Relationships between terms

- 1 Choose a word from the brackets to complete these statements so that they reflect positive citizenship.
- a When I choose an action, I need to be aware of the _____. (cost, time, consequences)
 - b Bullying and harassment are _____ acceptable behaviour. (sometimes, never, always)
 - c Being considerate means thinking of _____. (myself, others, my friend)
 - d By celebrating diversity, we help to make everyone feel _____. (included, excluded, precluded)

Active citizenship in action

- 2 Draw a line through the activities and attitudes that DO NOT reflect positive and active citizenship.

making poor decisions	withholding information	protecting the rights of others	understanding consequences
abiding by the law	valuing social justice	promoting inequalities	celebrating diversity
planning for the future	acting irresponsibly	making biased judgments	lack of involvement in activities
analysing information	contributing labour	ignoring all new trends	welcoming newcomers
being a volunteer	insulting people	forcing your views on others	being prepared to change

Addressing social issues through our own behaviour

- 3 The first column lists social issues. The second column lists positive actions that people can take to address each issue. Write numbers in the spaces to match each issue with an action. Note that alternative answers are possible.
- | | |
|---|---|
| 1 being annoyed at the ignorance of others | _____ participating in activities |
| 2 being misunderstood by others | _____ exercising the right to vote |
| 3 being worried about crime and its consequences | _____ explaining things about your life and beliefs to others |
| 4 feeling unsupported by others | _____ complying with the law |
| 5 worrying about loss of heritage and cultural identity | _____ educating people |
| 6 being lonely | _____ earning money |
| 7 having day-to-day needs to provide for | _____ contributing to the community, using your skills |
| 8 feeling insignificant as a participant in society | _____ socialising with others more regularly |
| 9 feeling powerless to affect the government | _____ supporting other people |
| 10 wanting to be more involved in community activities | _____ preserving and protecting cultural resources |

Shades of meaning

- 4 Write definitions for each of these words to show the distinctions between them. Some clues have been provided for you.

HINT: Noun – a word used to name a person, place, thing or quality. Verb – a word used to describe an action.

- affect (verb) _____
- effect (noun) _____
- quality (assessment) _____
- quantity (amount) _____
- independence (solo) _____
- interdependence (together) _____
- difference (opposite) _____
- diversity (various) _____



Answers

1 Australia's natural environment **1 a** F, **T b** T, F **c** F, T. **2** the Pacific Ocean, the Strzelecki Desert, the Coral Sea, the Great Sandy Desert, the Timor Sea, the Gibson Desert, the Tasman Sea, the Tanami Desert, the Arafura Sea, the Simpson Desert, the Indian Ocean, the Southern Ocean. **3 a** T, F, **b** T, F. **4 a** the Blue Mountains, **b** the Murray-Darling Basin, **c** the Southern Cross, **d** Fraser Island, **e** Mount Kosciuszko.

2 Mapping Australia **1** Check student answers against map. **2 a** 1100 km **b** 3000 km **c** 1600 km. **3** East coast areas of Vic, NSW and QLD, South-west coast of WA, South coast of SA, North coast of NT, South-east coast of Tas. **4** Temperate climates, regular rainfall, protective mountain ranges, close proximity to the sea, fresh water rivers, major cities (for infrastructure, access to transport, goods, services and employment opportunities).

3 Biodiversity in Australia **1** Answers will vary. Possible answers: banksia, bilby, cockatoo, dingo, eucalypt, echidna, golden wattle, hawk, ibis, jacaranda, kookaburra, mangrove, monitor, Norfolk pine, platypus, rosella, ti tree, Tasmanian devil, waratah, wedge tailed eagle. **2** Answers will vary. **3** (top row) fish, insect, reptile, arachnid (bottom row) bird, marsupial, invertebrate, mammal. **4 Plant:** bottlebrush, ghost gum, blackbutt, kelp, camphor laurel, water hyacinth, macadamia, mango. **Mammal/marsupial:** bandicoot, dolphin, dugong, flying fox, humpback, koala, possum, wombat. **Reptile:** lace monitor, death adder, tortoise, skink, goanna, turtle, king brown, crocodile. **Bird:** budgerigar, crow, currawong, penguin, lyrebird, tawny frogmouth, cassowary, emu. **Insect:** witchetty grub, mosquito, cicada, ladybird, dragonfly, praying mantis, Bogong moth, grasshopper. **Fish:** catfish, eel, lungfish, silver perch, Murray cod, barramundi, Mangrove Jack, flathead. **5, 6** Answers will vary.

4 Antarctica **1** Antarctic Circle, Southern Ocean, continent, location, square kilometres, ice, thickness, coldest, little, adapted, populated, freezing. **2** Blue whale (30 m), Fin whale (27 m), Southern Right whale (17 m), Humpback whale (16 m), Sperm whale (15 m), Minke whale (10 m), Southern Bottle Nose whale (9 m). **a** 425 m **b** 75 sperm whales **c** 54 tonnes **d** (40 tonnes x 2) + (80 tonnes x 2) + (46 tonnes x 3) = 378. $378 \div 7 = 54$ tonnes. **3** phytoplankton → krill → whales.

5 What geographers do **1 Physical:** determining the temperature inside an active volcano, forecasting the weather, measuring the height of an ocean tide, discovering how a plant is cross-pollinated. **Human:** investigating a factory's waste disposal practices, investigating unemployment in a particular city. **2** Answers will vary. **3** map: physical layout of the Earth or a specific location; thermometer: air or water temperature; photograph: the visual appearance of something; survey: a way to collect information or data about a particular topic; barometer: air pressure; seismograph: earthquake magnitude and other tectonic movements; tracking device: the movements of living organisms throughout their ecosystem. **4** Mind map structure: Total environment overarching all other elements, Physical Elements and Biological Elements under Total Environment; Rocks etc. under Physical Elements; Humans, Plants and Animals under Biological Elements.

6 Weather forecasting **1 a** swell, **b** north, **c** tide, **d** east, **e** knots, **f** west. **2 a** wind chill, **b** satellite, **c** coastal, **d** pollution, **e** temperature, **f** barometer, **g** pressure system, **h** UV index, **i** weather forecasting, **j** atmosphere. **3** air temperature, °C, knots, metres, %, time, minutes, earthquake magnitude. **4** Answers will vary. **5 a** barometer, rainfall, latitudes, swell; **b** outlook, knots, south, haze, east, trough, hectoPascals; **c** frost, temperature, earth, humidity; **d** tropic, coastal, low, westerly; **e** fog, gale force wind, degrees, scale, ecological;

f meteorologist, tornado, overcast; **g** front, tides, satellite; **h** pollution, northwest, total fire ban. **6** high pressure system: a series of widely spaced isobars forming an enclosed irregular shape with label 'H' in the centre; cold front: an arc with arrowheads spaced evenly apart; tropical cyclone: a series of concentric circular isobars with label 'TC' in centre.

7 Understanding latitude **1** Equator, zero, centre, line, angle, degrees, time, rotate, North. **2** Equator should be labelled in its correct position and curved horizontal lines drawn above and below in descending size. **3 Northern:** Greenland, Canada, Finland, England; **Southern:** Australia, South Africa, Peru, New Zealand. **4 a** Ecuador, Colombia, Venezuela; **b** Kenya, Uganda, Democratic Republic of Congo; **c** Borneo, Malaysia, Indonesia.

8 Understanding longitude **1** imaginary, curved, single, meridian, reference, coordinates, measured, London, longitude, west, half, circumference. **2** Greenwich Meridian marked as a central line passing through southern England, and lines drawn fanning out from the North to the South Pole. **3 Eastern:** Egypt, Turkey, India, Saudi Arabia, Japan, Austria. **Western:** Morocco, Argentina, Ireland, USA.

9 Famous places around the world **1 a** Grand Canyon **b** Great Barrier Reef **c** Rio de Janeiro Harbour **d** Mount Everest **e** Aurora Borealis **f** Paricutin Volcano **g** Victoria Falls. **2 a** Amazon **b** Colorado **c** Murray-Darling **d** Ganges **e** Jordan **f** Nile **g** Rhine **h** Rio Grande **i** St Lawrence **j** Seine **k** Shannon **l** Thames **m** Volga **n** Yangtze **o** Zambezi. **3 a** Switzerland, **b** Scotland, **c** Australia, **d** Russia. **4 a** Lake Superior **b** Lake Victoria **c** Lake Malawi **d** Caspian Sea. **5 a** White Cliffs of Dover, **b** Lake Victoria, **c** Dublin, **d** Kyoto, **e** Mount Vesuvius, **f** Berlin, **g** Johannesburg. **6** Answers will vary.

10 The four spheres **1 Atmosphere:** wind, air pressure, cirrus clouds, air, meteorology, ozone, oxygen, gases, stratosphere, temperature, carbon dioxide. **Biosphere:** biodiversity, reproduction, ecosystem, vegetation, lichen, consumer, habitats, predator, food webs, rainforest, photosynthesis. **Hydrosphere:** oceans, rivers, waves, estuary, tides, hail, dew, precipitation, tsunami, snow, icebergs. **Lithosphere:** crust, mountains, deserts, soil, topography, mantle, valleys, geology, fault, plate tectonics, plateau. **2 a** A, **b** A, **c** L, **d** B, **e** L, **f** H, **g** A, **h** H, **i** L, **j** L, **k** B, **l** H. **3 a** gases, **b** living, **c** water, **d** crust. **4 a** **4 b** **7 c** hydrosphere **5 a** atmosphere, atmos, amos, amostquoi, mostquoi, mosquito. **b** biosphere, biosphere, bofere, bfeer, beef. **c** hydrosphere, hydrphre, hydrphreate, hydrreate, hydrate, hydration, dehydration. **d** lithosphere, lithoshere, shire, hotel.

11 Nature's fury **1** tornado, twister, air; landslide, landslip, earth; tsunami, tidal wave, water; earthquake, tremor, earth; cyclone, hurricane, air; blizzard, white out, air; fog, mist, air; flood, inundation, water; sandstorm, willy willy, air; bushfire, blaze, fire. **2 b, d, f, h, j, c, e, a, g, i.** **3** Answers will vary. **4** avalanche: the rapid downhill movement of a mass of unstable snow and ice; thunderstorm: a storm resulting from strong rising air currents, heavy rain or hail and thunder and lightning; volcano: a mountain or hill formed from a vent in the earth's crust from which lava, ash and other materials are violently ejected. **5 b, f, d, g, a, e, c.** **6, 7** Answers will vary. **8** Possible answers: cuts, abrasions, headache, broken bone, bullet wound, fever, diabetic episode, implement wound, hypothermia, infection, bite wound.

12 The structure of the Earth **1:** 1 core 2 mantle 3 crust 4 ocean 5 continent. **2:** 2 hill, 4 plain, 1 mountain, 3 plateau. **3** Drawings should show: mountain (steep slopes, with jagged points at the skyline); plateau (elevated with a fairly flat surface); hill (rounded with gentle slopes); plain (low in



elevation, flat or very gently sloping). **4** super continent, seismic activity, faultline, Pacific Ring of Fire, Richter scale, continental drift, plate tectonics, sea bed.

5 Nouns: cracks, surface, relief, lithosphere, collision, fissure, magma, tsunami, debris; **Adjectives:** forceful, subterranean, layered, oceanic, massive. **Definitions:** Answers will vary.

13 Movement of the Earth's crust **1 a** weathering, **b** water erosion, **c** wind erosion, **d** ice erosion, **e** folding, **f** faulting **g** earthquakes, **h** volcanic activity, **i** landslides, **j** rock falls, **k** mud flows, **l** avalanches. **2** Answers will vary. **3** The graph should present seismic events along the x (horizontal) axis and fatalities along the y (vertical) axis. Scale should be from zero to 300,000, in increments of 20,000 to 30,000.

14 Volcanoes **1 a** crater, extinct, dormant, fissure, pyroclastic, active, vulcanology. **b** granite, steam, smoke, pumice, ash, lava, mud, basalt, gases. **c** collapse, explode, eruption, flowing, blast. **2 a** magma, lava. **b** pressure, chemically. **c** sediments, layers. **3 Igneous:** granite, basalt, pumice, obsidian. **Metamorphic:** gneiss, marble, slate, quartzite. **Sedimentary:** dolomite, sandstone, limestone, chalk. **4** **1** lava, **2** magma chamber, **3** ash layer, **4** vent. **5 a** Smoke is produced from substances burning and producing carbon (ash) whereas steam is water vapour. **b** Magma is the molten material found inside the Earth's mantle; when it breaches the surface it is called lava. **c** To erupt is to explode, whereas to eject means to expel or throw out forcefully. **6** Etna, Sicily, N; Vesuvius, Italy N; St Helens, USA, N; Krakatoa, Indonesia, S; Fuji, Japan, N; Vulcan, Papua New Guinea, S.

15 Mountains of the world **1 a** Mount Everest, **b** The Himalayas, **c** Nepal, **d** Edmund Hillary, **e** New Zealand. **2 a** K2, **b** Mount Kilimanjaro, **c** Mount McKinley, **d** Mount Cook, **e** Mount Fuji, **f** Ben Nevis, **g** Mount Aconcagua. **3** tsunami, **4** vulcanology, **5** Diagrams should show rock material being broken in masses (faulting) and being crumpled and curved but unbroken (folding).

16 Minerals of the Earth **1** Puzzle: see solution on page 48. **2 Metals:** iron, zinc, lead, silver, tin, gold, magnesium, titanium, nickel, copper. **Stones:** diamond, jade, agate, emerald, jasper, peridot, quartz, sapphire, flint, amethyst, sandstone, ruby, alabaster, topaz, marble, beryl, onyx, limestone, granite, garnet, opal, turquoise, lapis lazuli.

17 Oceans **1** Pacific, Atlantic, Indian. **2** Mediterranean, North, Black, Dead. **3** Coral Sea, Tasman Sea, Southern Ocean. **4** canopy, ozone. **5** mariner, marina. **6 a** frigate, fishing trawler, dinghy. **b** cargo ship, tugboat, lifeboat. **c** passenger liner, patrol boat, canoe. **d** destroyer, houseboat, ski boat. **7** measurement, mile, 1.15, convert, multiply. **8** 15 knots x 1.15 = 17.25 then multiply by 1.609344 = 27.76 km. **9 a** oil drilling, **b** whaling, **c** over-fishing, **d** nuclear testing, **e** Indigenous rights. **10 Fishing:** driftnets, long lines, catch limits. **Mining:** natural gas reserves, oil platforms, minerals. **Tourism:** water sports, snorkelling, whale watching. **Trade:** exports, merchant shipping, cargo ships. **Immigration:** refugees, border protection, illegal entry. **11** **1** ocean, **2** lake, **3** evaporation, **4** condensation, **5** transpiration, **6** precipitation, **7** land, **8** infiltration, **9** groundwater, **10** run-off.

18 The Sun **1 a** solar powered, **b** radiant heat, **c** anti-nuclear protests, **d** water pollution, **e** the Earth's atmosphere, **f** energy efficiency, **g** information technology, **h** natural processes, **i** climate change, **j** the four seasons, **k** carbon dioxide emissions, **l** conserving fossil fuels. **2 a** greenhouse effect, **b** climate change, **c** global warming, **d** solar energy, **e** ultraviolet light. **3 & 4** Answers are approximate. **3 a** -13°C, **b** 16°C, **c** 38°C. **4 a** 32°F, **b** 96°F, **c** 60°F.

19 The ozone layer **1** gas, rays, ultraviolet, less, destruction or depletion, layer, diseases, hole, marine, food. **2** phytoplankton: microscopic single-celled organisms that use light to produce food through photosynthesis; ultraviolet light: the rays of sunlight that emanate from beyond the visible violet end of the spectrum, which are harmful to some organisms; polluting waste products and substances negatively affect the natural environment and living things; CFCs (chlorofluorocarbons): chemicals used in certain propellant-based products, which are harmful to the ozone layer and the environment. **3** spray cans, air conditioners, foam plastics. **4** sunlight → phytoplankton → krill → whale. **5** Answers could include reducing energy consumption, using pump sprays instead of CFC-reliant products, using environmentally clean transportation and educating the public.

20 Clouds **1 a** stratus, **b** cumulus. **2** cirrus, cumulus, nimbus, stratus. **3 a** liquid, **b** cycle, **c** hydrosphere, **4** vapour, droplets, suspended, clouds, surface, precipitation, increase, heavy, supported. **5** **1** cirrus, **2** nimbus (or cumulonimbus), **3** cumulus, **4** stratus. **6** Cirrocumulus are fluffy, fairly small and high in the atmosphere; cirrostratus are wispy and layered; altostratus are grey formless clouds (overcast); altocumulus are grey, puffy-shaped clouds; nimbostratus are low, dark grey sheets of rain clouds; stratocumulus are grey-white elongated sheet-type clouds. **7 a** When cold air pushes under a mass of warm air, the lighter warm air rises and condensation occurs. **b** This process of air movement creates cumulonimbus clouds in the atmosphere. **c** These large, dense clouds gather to create thunderstorms, with heavy rainfall and sometimes hail. **8** Possible answers: **a** A cloud is a visible mass of tiny water droplets floating in the atmosphere. **b** The tiny droplets and crystals in the clouds reflect the sunlight, making them appear to be white. **c** Water vapour condenses so rapidly that it freezes and falls as hail.

21 Desert environments **1 a** dryness, **b** rainfall, **c** friction, **d** dumping, **e** wearing away, **f** flora, **g** lack. **2** Answers will vary. **3 a** Dig down deep into the sand until water begins to pool at the bottom. **b** Dig up deep tree roots, cut them open and collect their stored water. **c** At night time, collect the condensation from the moist cold air using a piece of fabric. **d** Find and follow any birds or animals to a water source. **4 h, a, d, g, i, c, b, e, f.** **5** Answers will vary. **6 a** The windward side of a sand dune is exposed to the wind, and the leeward side is sheltered from the wind. **b** Deposition is the placing down of material being transported by erosion, whereas deflation is the removal of particles by the wind. **c** Rock material is worn into smaller particles (weathering) and the particles are transported away (erosion).

22 Grasslands **1 a** One type of tropical grassland is the savanna. **b** The grasslands of North America are known as prairies. **c** Summer is the wettest season in tropical savannas. **d** Bison, buffalo and deer once grazed on the American prairies. **e** Grass fires often occur at the end of winter. **f** Lions are the biggest carnivores of the African grasslands. **g** Grasslands provide a good home to herbivores. **2 a** temperate, **b** rainfall, **c** Tropics, **d** treeless, **e** continent, **f** relatively, **g** on. **3 a** North America, **b** Africa, **c** Australia. **4 a** Africa, **b** Australia, **c** North America. **5 a** Carnivores kill and eat herbivores. Two types of carnivores are predators and scavengers. **b** Many grasslands have now been made into farmland. Kruger National Park is a grassland in South Africa. **c** The process of grasslands turning into deserts is called desertification. One of the main problems threatening the grasslands is overgrazing. **d** Hunting and poaching have caused a decrease in the numbers of some species. Stricter laws are needed to help protect the grassland animals for future generations. **6** savannas, pampas, plains, steppes, prairies, veldts, campos.

23 Wetlands **1 a** bogs and fens, **b** swamps and marshes, **c** estuaries and tidal flats, **d** mudflats and salt marshes. **2 b** habitat, **c** mangroves, **d** Everglades, **e** Kakadu. **3** Diagram should include: vertical arrows pointing upwards to show evaporation and transpiration; horizontal arrows to show condensation in the atmosphere above the points of evaporation and transpiration; vertical arrows pointing downwards to show precipitation falling on the land; slanted horizontal arrows pointing towards the lake and / or ocean to show run-off flowing and / or collecting back in the wetland. **4** ducks, water rats, fish, alligators, otters, beavers, frogs, herons, shellfish, platypuses, geese, swans, bacteria, mosquitoes and snakes.

24 Rainforest environments **1 a** tropical, **b** canopy, **c** ecosystem, **d** habitat, **e** photosynthesis, **f** species, **g** pollination, **h** moss, **i** ferns, **j** humidity. **2** Answers could include: plants convert sunlight into food and oxygen; sunlight is converted to sugars; green leaves absorb and process sunlight; leaf cells contain chlorophyll for use in photosynthesis, which requires light and water; plants consume water, sunlight and carbon dioxide and produce sugars and oxygen from them. **3** Incorrect statements: **Ground Level** is very sunny and warm; is a good place for photosynthesis. **Lower Storey** has many green plants; has lots of light available; is dry and dusty. **Upper Storey** is located near the ground; is full of moss and lichens. **4 a** Equator, **b** pollinate, **c** water, **d** deforestation. **5** Australia, Fiji, Indonesia, Congo, Ecuador and Venezuela.

25 Mountain environments **1** 10, 5, 8, 9, 25, 13, 11, 17, 1, 21, 29, 22, 23, 24, 12, 7, 4, 16, 30, 6, 20, 27, 2, 19, 15, 3, 28, 26, 14, 18. **2 a** 3, 2, 1. **b** 2, 1, 3. **c** 3, 2, 1. **d** 1, 3, 2. **e** 1, 3, 2. **f** 2, 1, 3. **3** Puzzle: see solution below right. **4** HIMALAYAS, ANDES, ROCKIES, URALS, ATLAS, CAUCASUS, ALPS.

26 Marine environments **1 a** What small animals are responsible for forming coral reefs? **b** What do these tiny animals eat? **c** Coral is made from which substance? **d** What living things do coral reefs help support? **e** Do coral reefs grow quickly or slowly? **f** In which ocean are most of the world's coral reefs? **g** What is one human activity that could threaten a coral reef? **2 a** T, **b** F, **c** F, **d** T. **3** range, species, grow, maintain, depend on, survival, predators, large, smaller, provide, mammals, ecosystem, damaged, marine, endangered. **4 a** calcium carbonate, **b** symbiotic relationship, **c** fragile ecosystem, **d** box jellyfish, **e** primitive organism, **f** polyp colony, **g** sea anemone, **h** growth rate, **i** food chain, **j** marine life, **k** limestone skeleton. **5 a** algae and coral polyps, **b** large fish, **c** plankton, **d** smaller fish. **6 a** algae → small fish → large fish → sharks, **b** seaweed → tiny fish → crabs → humans.

27 Endangered species **1 a** Indonesia, List J; **b** China, List F; **c** Sumatra, List G; **d** Antarctica, List C; **e** Congo, List I; **f** Australia, List D; **g** North America, List B; **h** Lapland, List E; **i** Kenya, List H; **j** Madagascar, List A. **2 a** polar bear, **b** white rhinoceros, **c** blue whale, **d** tree kangaroo (found in Australia). **3** Answers will vary.

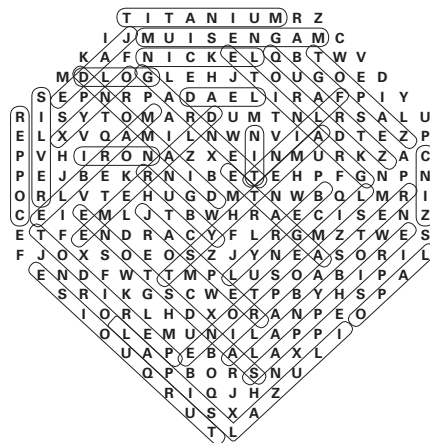
28 Bushfires **1 a** natural processes – spontaneous combustion, lightning strikes; vegetation – grass, trees and leaf matter; ignited – caught alight. **b** conditions – weather; unpredictable – direction constantly changing or impossible to predict. **c** evacuated – asked to leave the area for their own safety; residential areas – sites where numerous homes are clustered together; **d** fauna – animal life; habitats – the natural environment a species calls home. **e** arson – the deliberate lighting of fires; devastating – very damaging and destructive. **f** back-burning – lighting and management of controlled fires to eliminate fuel for bushfires; potential fuel – dry leaf matter and other flammable items. **2 a** 1 arson, 2 fire fighting. **b** 1 fire,

2 evacuation. **c** 1 lightning strike, 2 scrub fire. **d** 1 igniting, 2 waterbombing. **3 a** Fires produce a lot of smoke, heat and ash. **b** They present a danger to fire fighters and can result in injuries and loss of life. **c** Fire fighting costs a lot of time, manpower and resources. **d** Bushfires sometimes force people to leave their homes. (Variations acceptable if grammatically correct.)

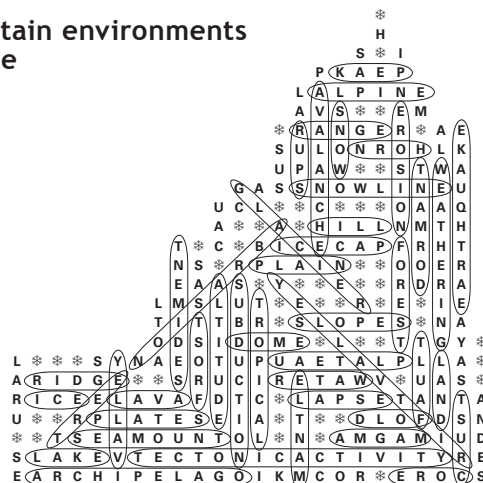
29 The built environment **1** industrial – factories, sewage plant, gasworks, power grid; recreational – parks, cinema, amusement parks, zoo; commercial – shops, markets, shopping mall; financial – stock exchange, insurance offices, banks; residential – houses, flats, apartments; educational – colleges, university, schools; religious – mosques, cathedral, churches; waterfront – fisherman's wharf, marina, docks; judicial – police headquarters, courthouse; administrative – government offices, social services offices. **2 a** North, **b** Brunton Avenue, **c** Five, **d** Victoria and Elizabeth, **e** Myer Music Bowl, National Tennis Centre, Melbourne Cricket Ground. **f** Parliament House. **3** Answers will vary.

30 Active citizenship **1 a** consequences, **b** never, **c** others, **d** included. **2** Draw a line through: making poor decisions, withholding information, promoting inequalities, acting irresponsibly, making biased judgments, lack of involvement in activities, ignoring all new trends, insulting people, forcing your views on others. **3** 8, 9, 2, 3, 1, 7, 10, 6, 4, 5. **4** affect – to bring about; effect – a consequence; quality – the relative worth of something; quantity – the amount of something; independence – acting individually; interdependence – depending on yourself and others; difference – lack of similarity or opposition between things; diversity – variety among things.

16 Minerals of the Earth puzzle



25 Mountain environments puzzle



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